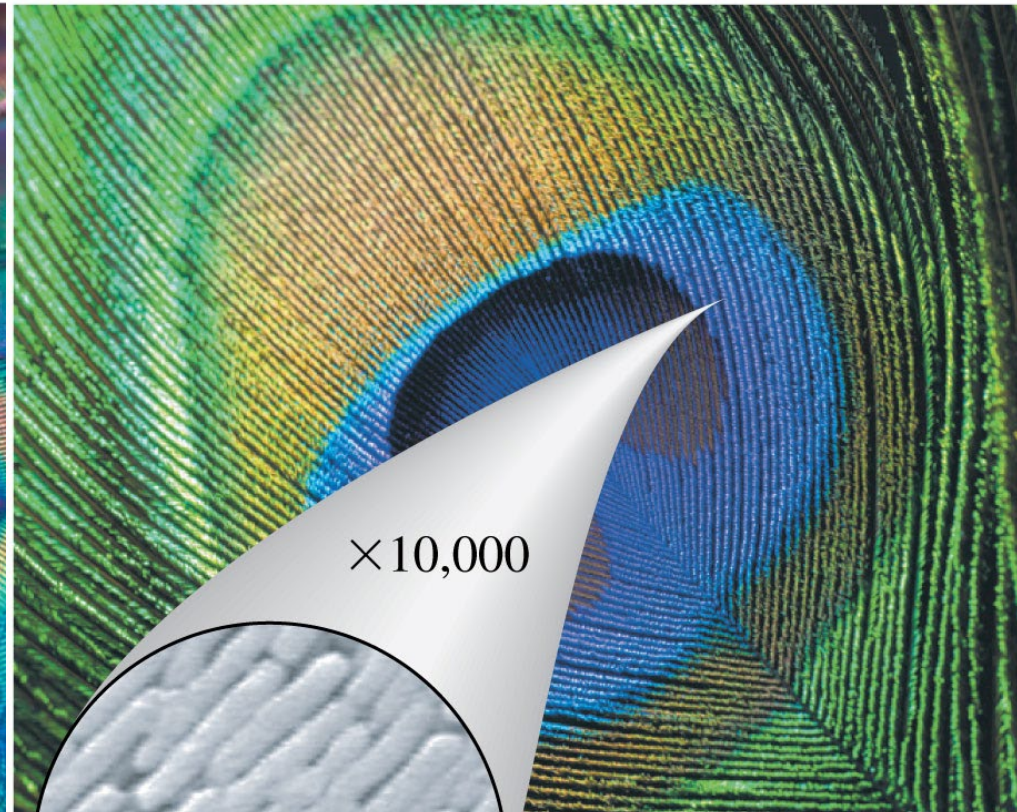
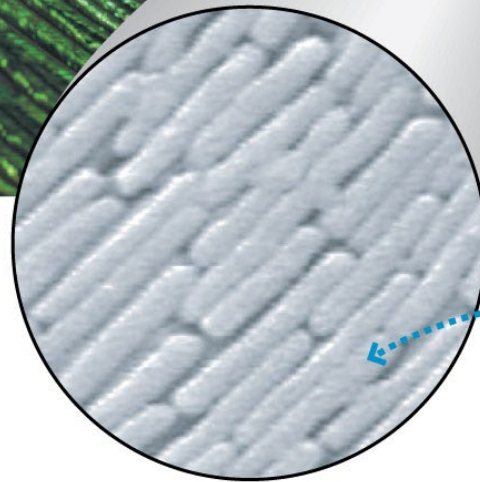


WAVE OPTICS



$\times 10,000$



Parallel rods of melanin form a diffraction grating.

Objectives

- To introduce the *wave model of light*.
- To understand how & why *interference of light* occurs.
- To understand the interference patterns of *double slits & diffraction gratings*.
- To study the inevitable spreading of waves due to *diffraction*.

Models of Light

- **Wave model:** Under many circumstances, light exhibits the same behavior as sound or water waves. The study of light as a wave is called *wave (physical) optics*.
- **Ray model:** The properties of prisms, mirrors, and lenses are best understood in terms of *light rays*. The ray model is the basis of *ray (geometrical) optics*.
- **Photon model:** In the quantum world, light behaves like neither a wave nor a particle. Instead, light consists of *photons* that have both wave-like and particle-like properties. This is the *quantum theory* of light.

$$\lim_{N \gg 1} \{\text{quantum optics}\} = \{\text{physical optics}\}$$

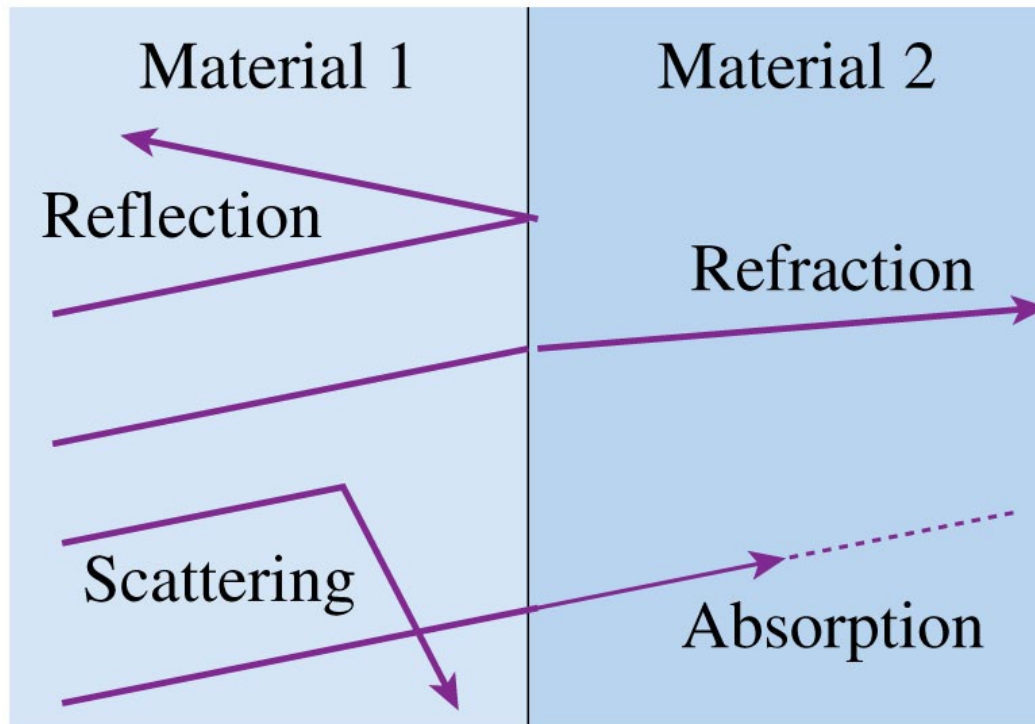
$$\lim_{\lambda \rightarrow 0} \{\text{physical optics}\} = \{\text{geometrical optics}\}$$

The Ray Model of Light

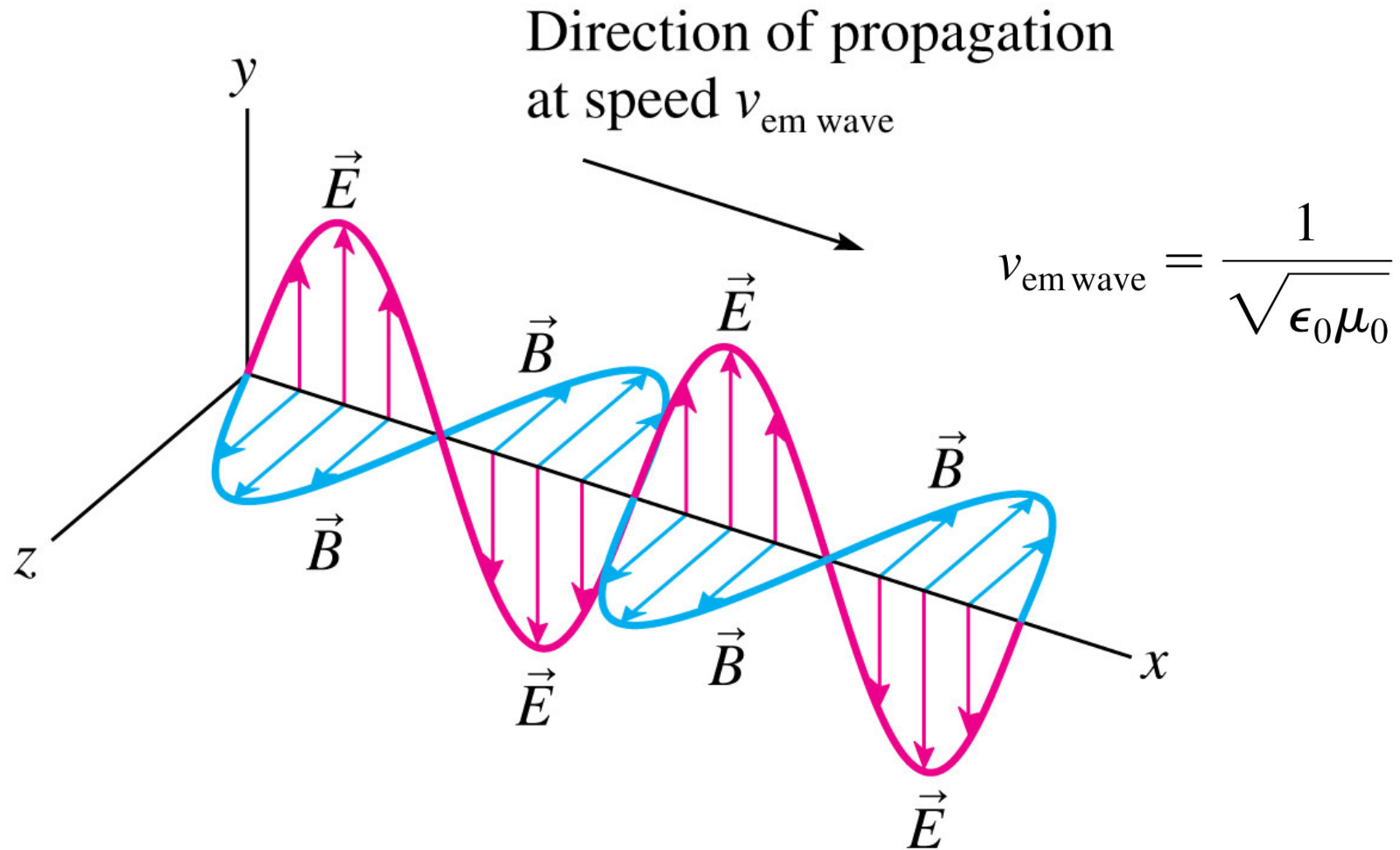
Light interacts with matter:

At an **interface** between two materials, light can be **reflected** and/or **refracted**.

Within a material, light can be **scattered** and/or **absorbed**.



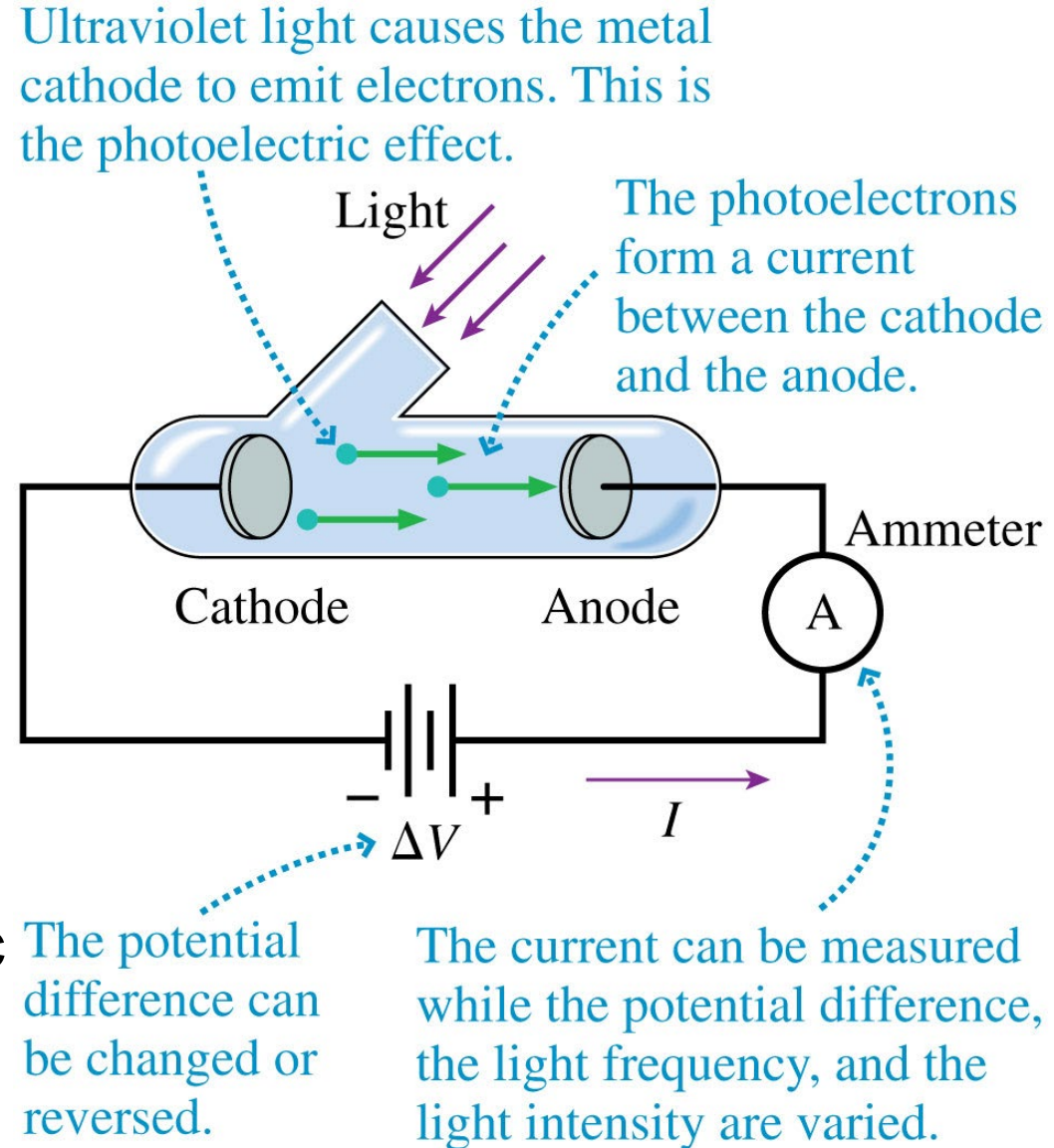
Maxwell's Theory of Electromagnetic Waves



- This is an **electromagnetic wave**.

The Photoelectric Effect

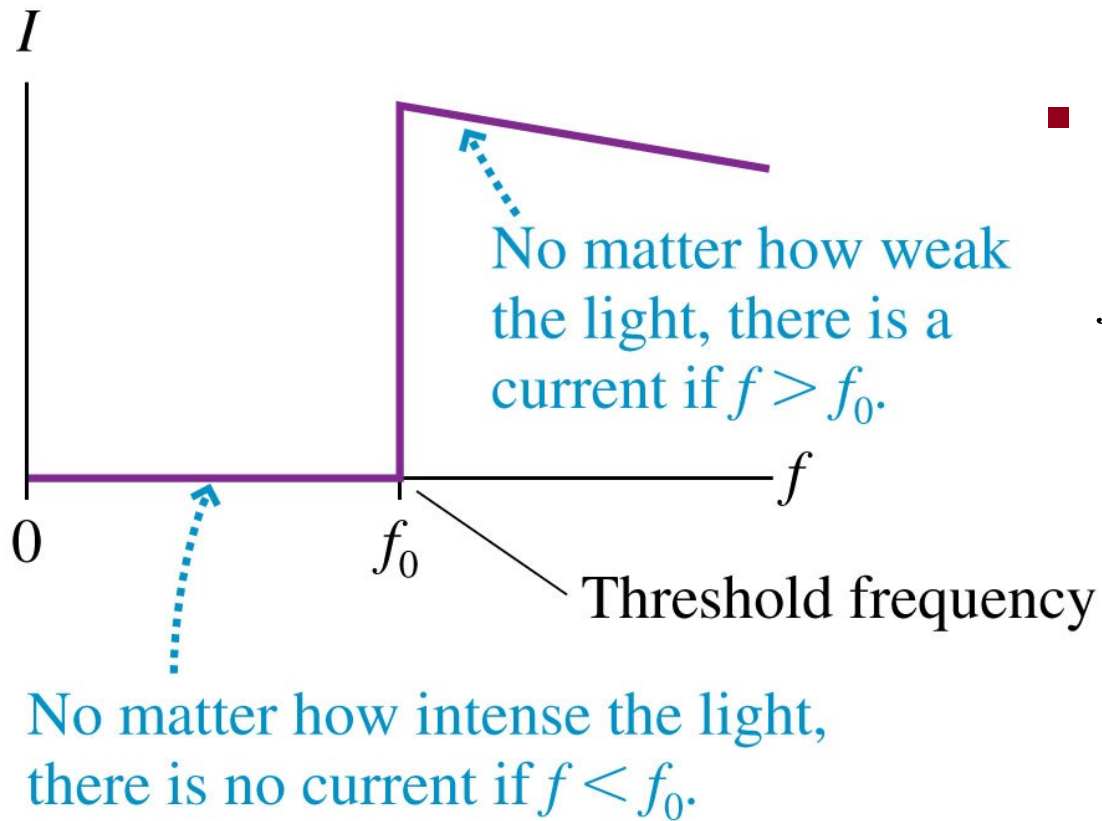
- Around 1900, Phillip Lenard built an apparatus which produced an electric current when ultraviolet light was shining on the cathode.



- This is the **photoelectric effect**.

Characteristics of the Photoelectric Effect

- Photoelectrons are emitted *only* if the light frequency f exceeds a **threshold frequency** f_0 .

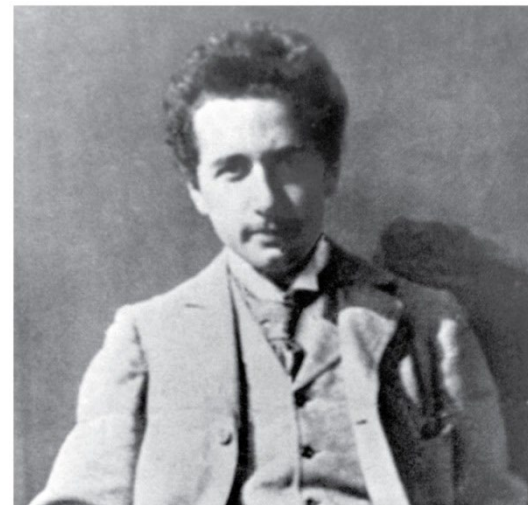


- The value of the threshold frequency f_0 depends on the type of metal from which the cathode is made.

Quantization of Light

- Around 1900, Max Planck was able to predict the spectrum of blackbody radiation by assuming that atoms oscillating with frequency f could only have specific energy values $E = 0, hf, 2hf, 3hf, \dots$, where **Planck's constant** is $h = 6.63 \times 10^{-34} \text{ J s}$.
- In 1905, Albert Einstein suggested that the **electromagnetic radiation itself is quantized!**
- Einstein called each packet of energy a **light quantum**, with energy:

$$E = hf$$



Before:



Work function E_0

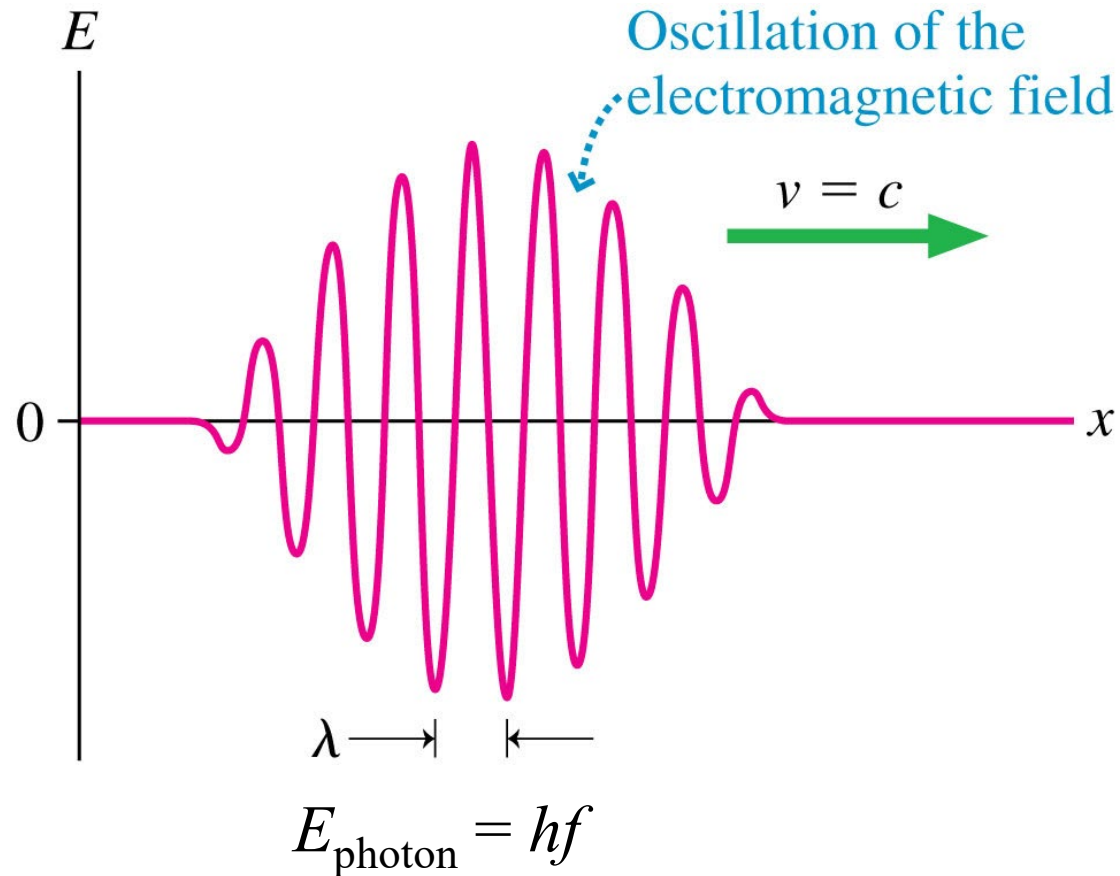
One quantum of light with energy $E = hf \geq E_0$

After:



A single electron has absorbed the entire energy of the light quantum and has escaped.

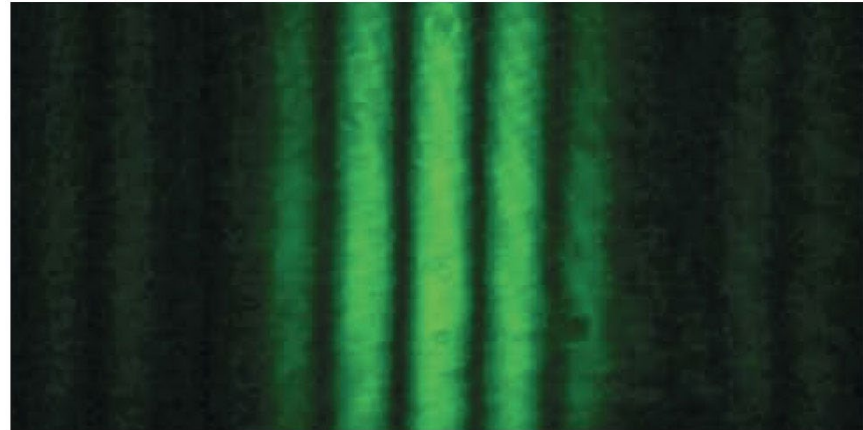
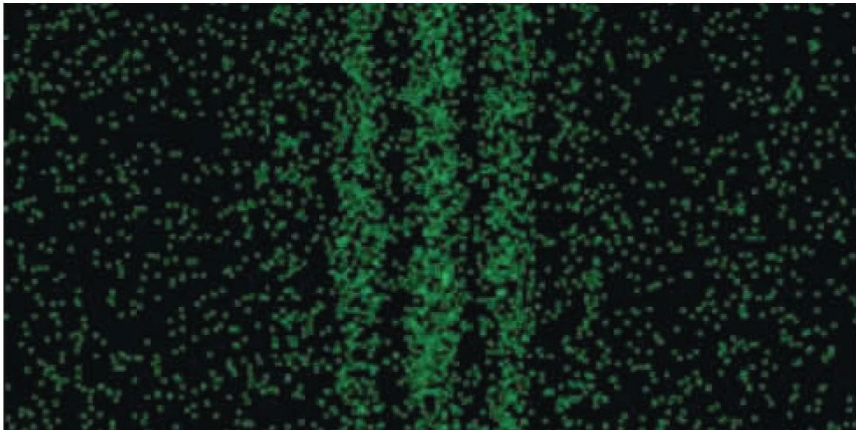
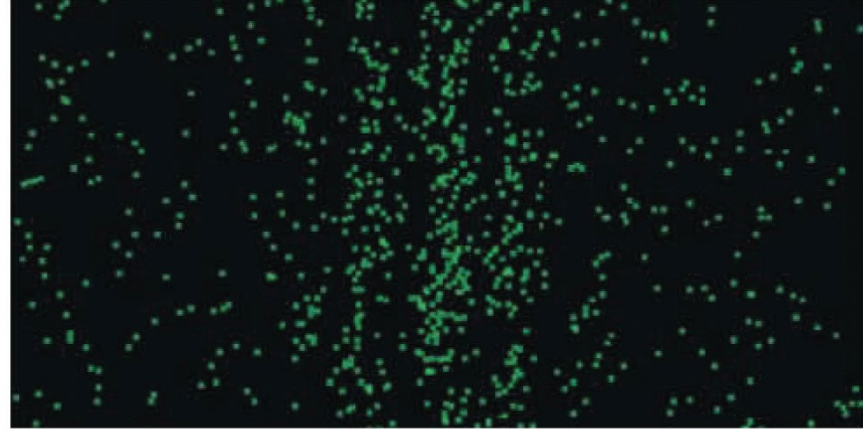
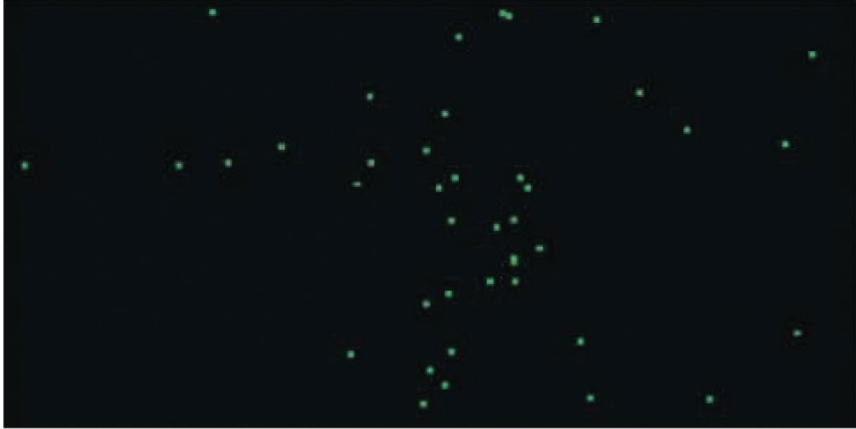
The Photon Model of Light



- **Photons** are sometimes visualized as **wave packets**.

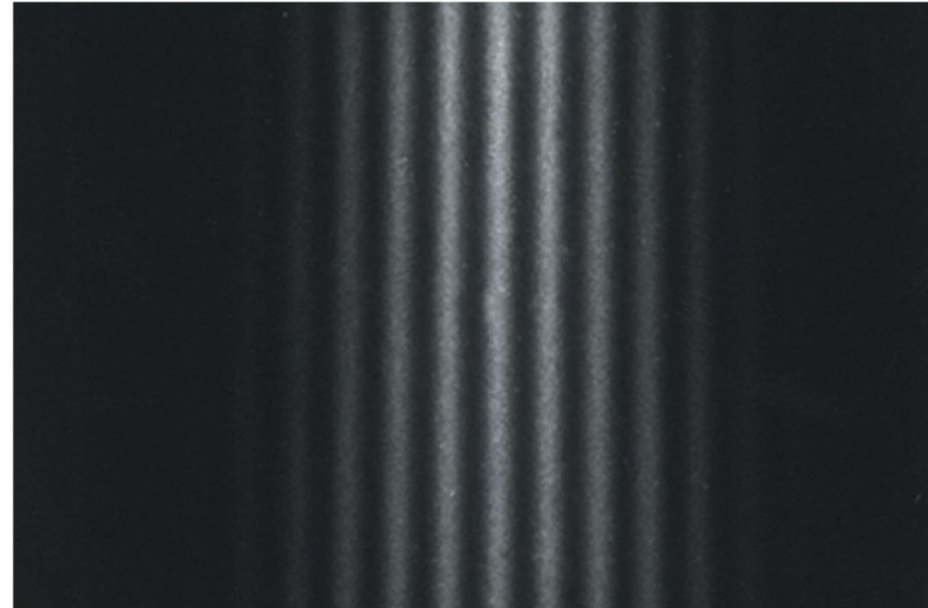
- The electromagnetic wave shown in the figure has a wavelength and a frequency, yet it is also discrete and fairly localized.

Double-Slit Experiment with *Single* Photons



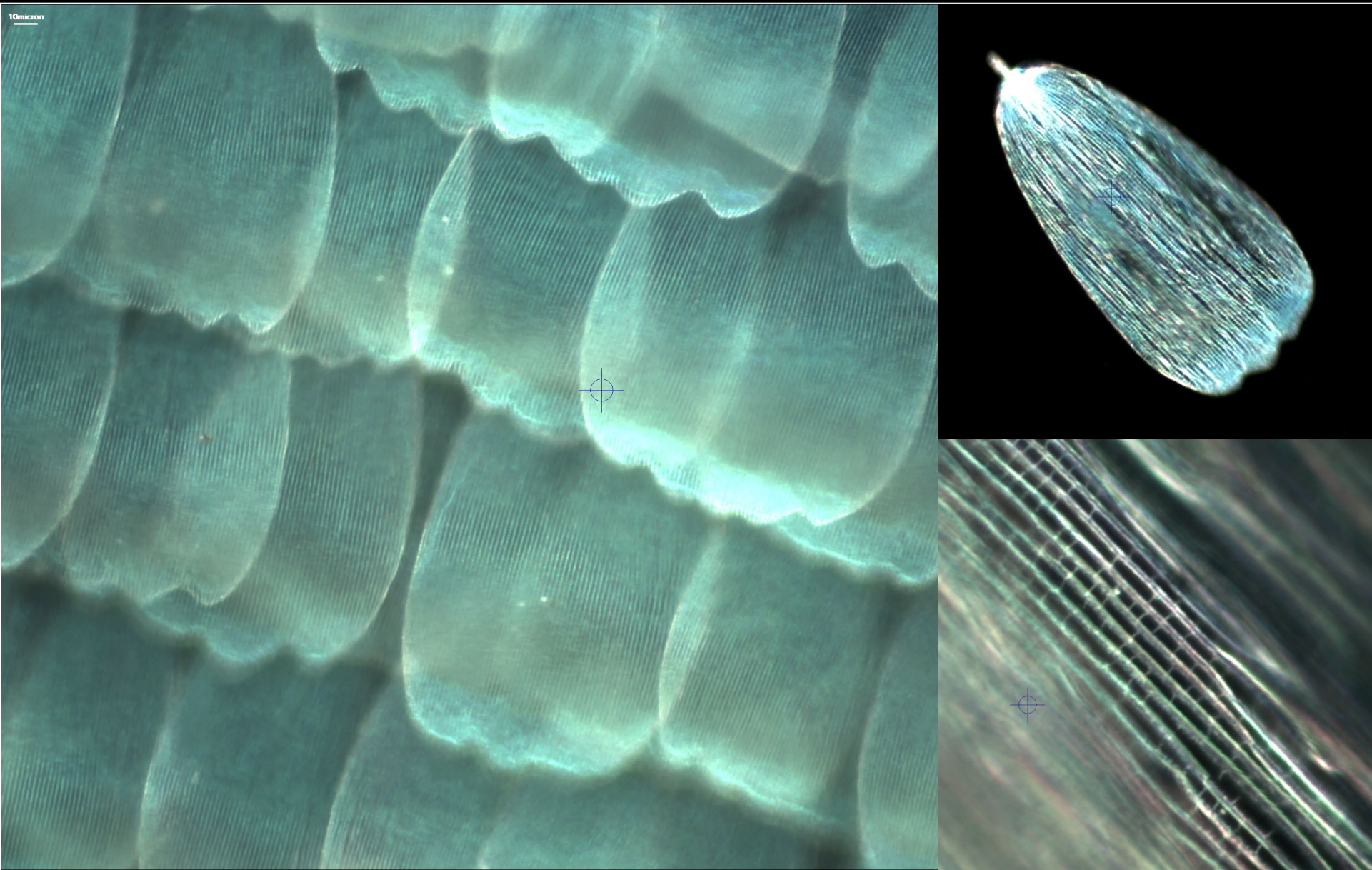
Matter Waves

- The image to the right shows the intensity pattern recorded after 50 keV **electrons** passed through two slits separated by $1.0\text{ }\mu\text{m}$.



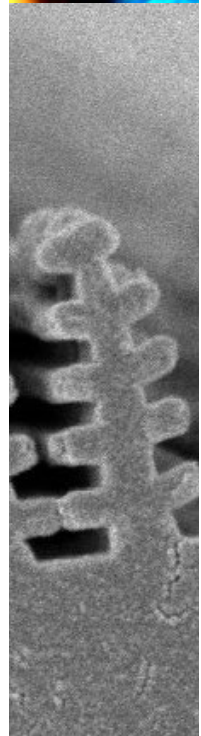
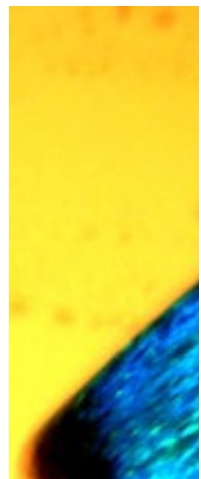
- The pattern is clearly a double-slit interference pattern.
- Electrons, neutrons, atoms, and even molecules exhibit all the behavior we associate with waves.

Morpho Butterfly Wing Scales

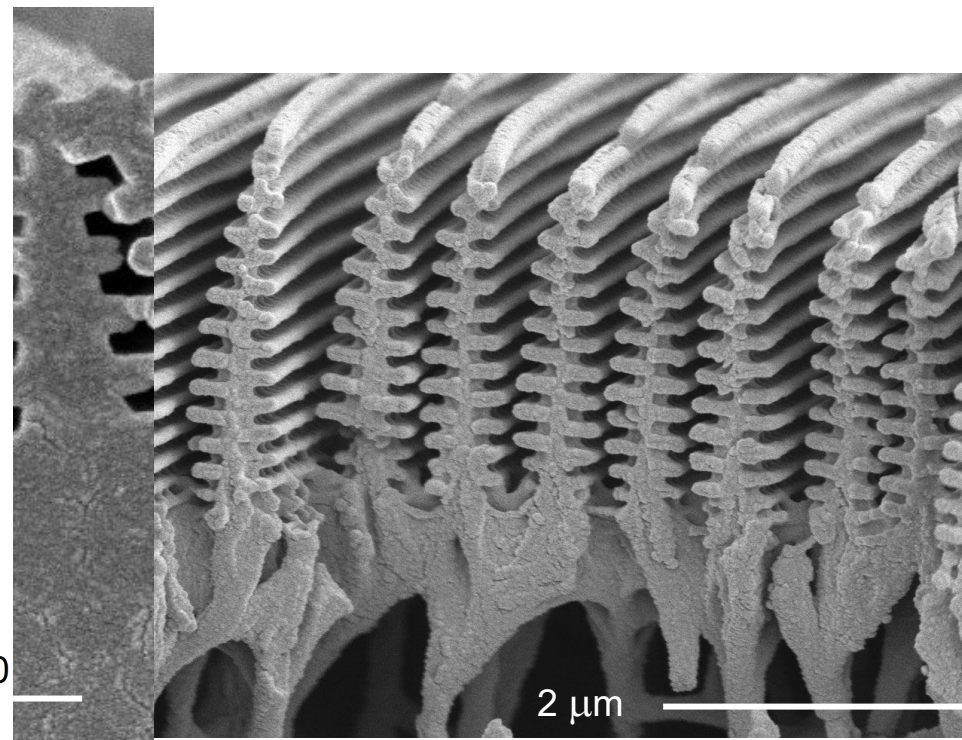
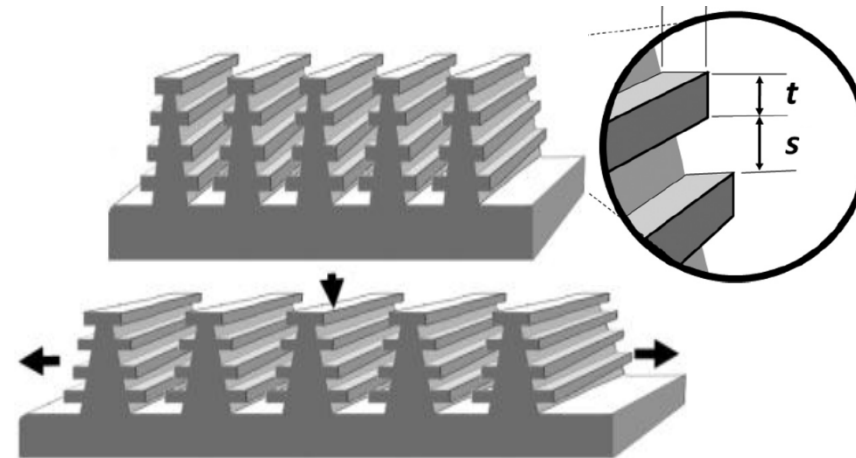
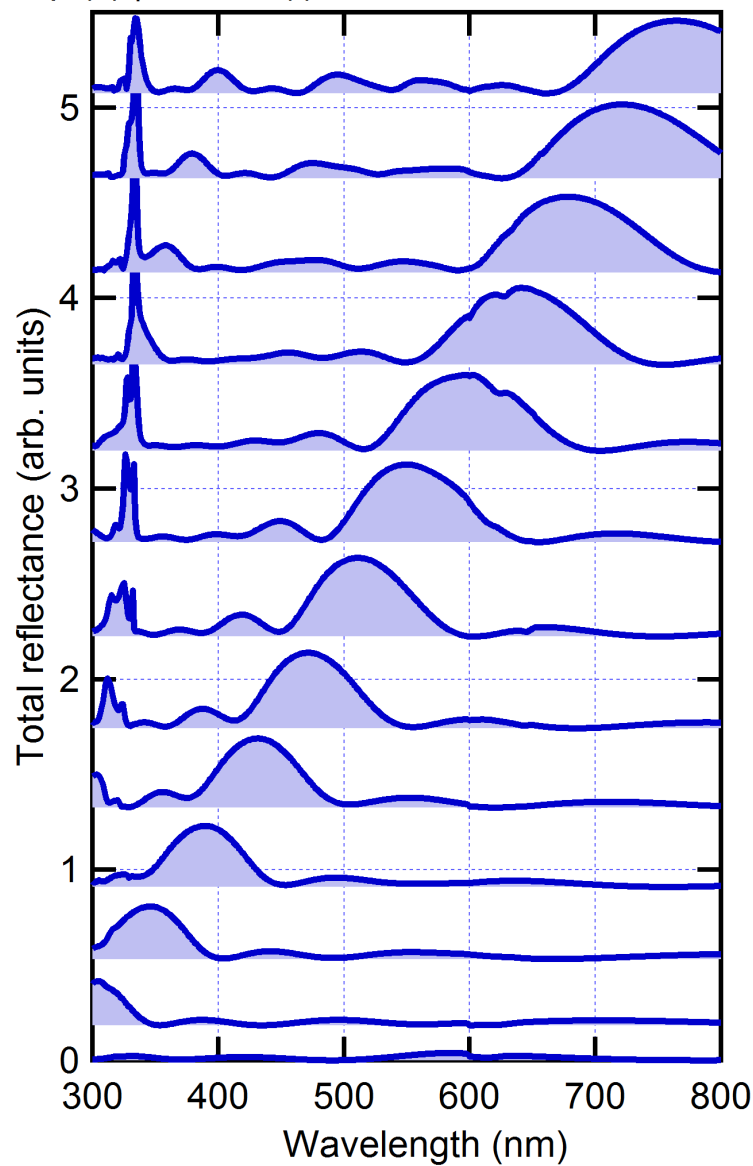




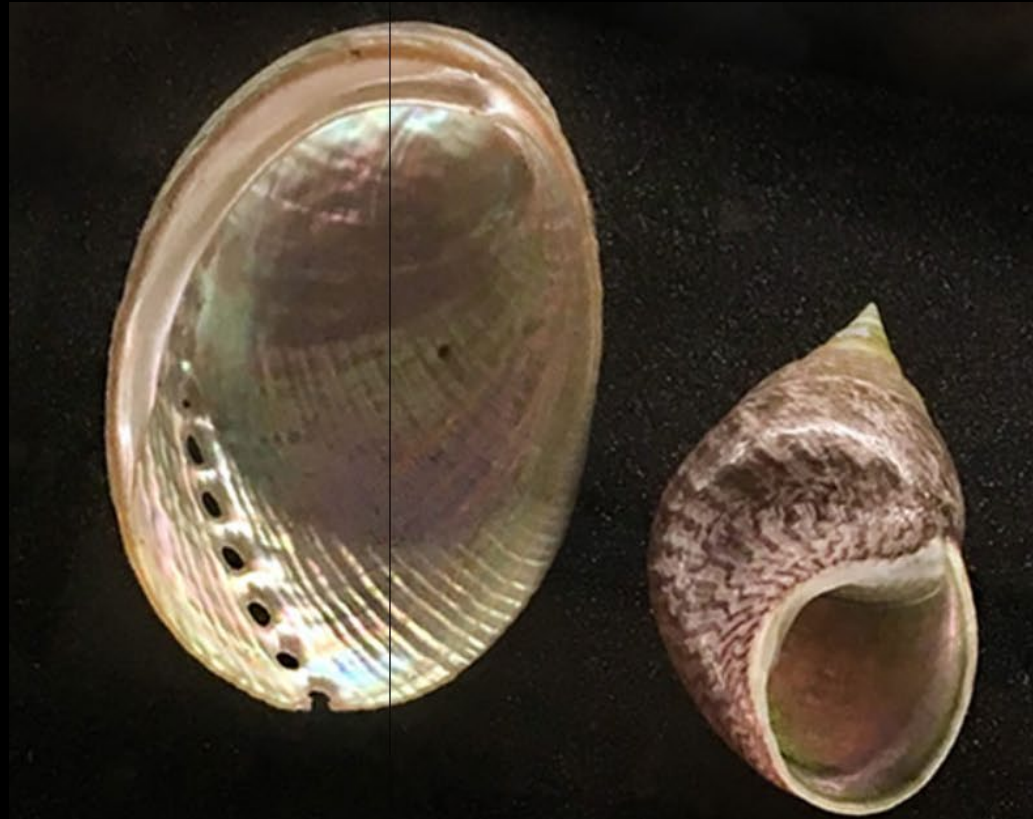
Tunable Coloration using *Morpho*-Inspired Polymer Nanostructures



Gap (s) | Branch (t) thickness: 50 $\uparrow$ 170 nm



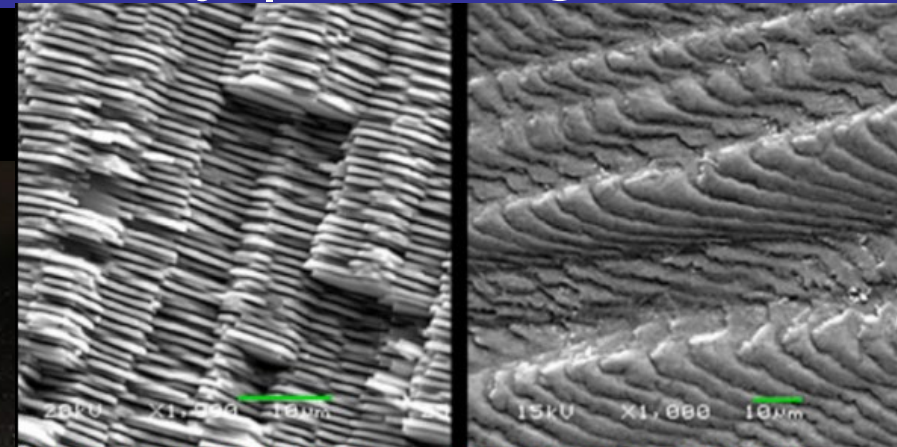
OPoD: Optics Picture of the Day | *Shining Shells*



Shining Shells

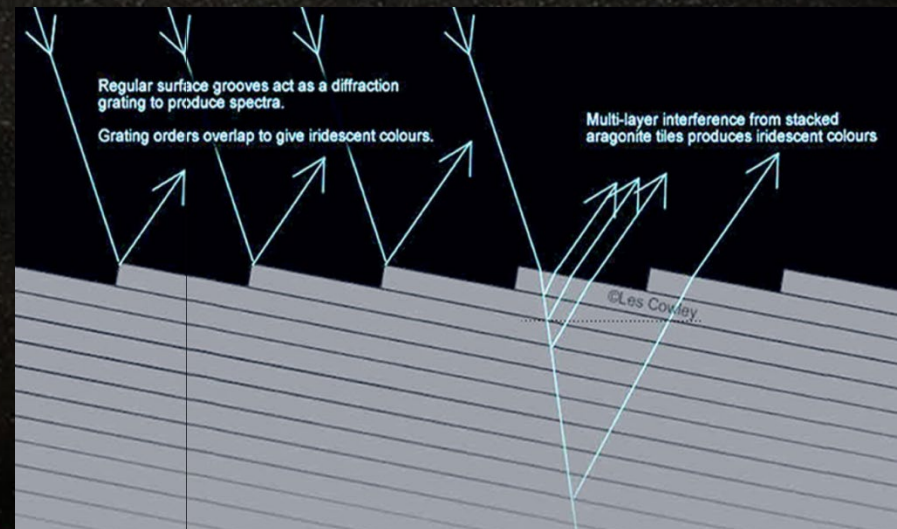
I came across these gem-like mollusc shells a few days ago in the [Museum of Zoology](#) in Cambridge (UK). This freshly reopened museum is delightful, brain stirring and a must. The image does little justice to the the shells, they shone colours brighter than the gems of a jeweller's window. Their secret lies in the structure of their lining of nacre, a miracle of bio-engineering.

Image ©Les Cowley

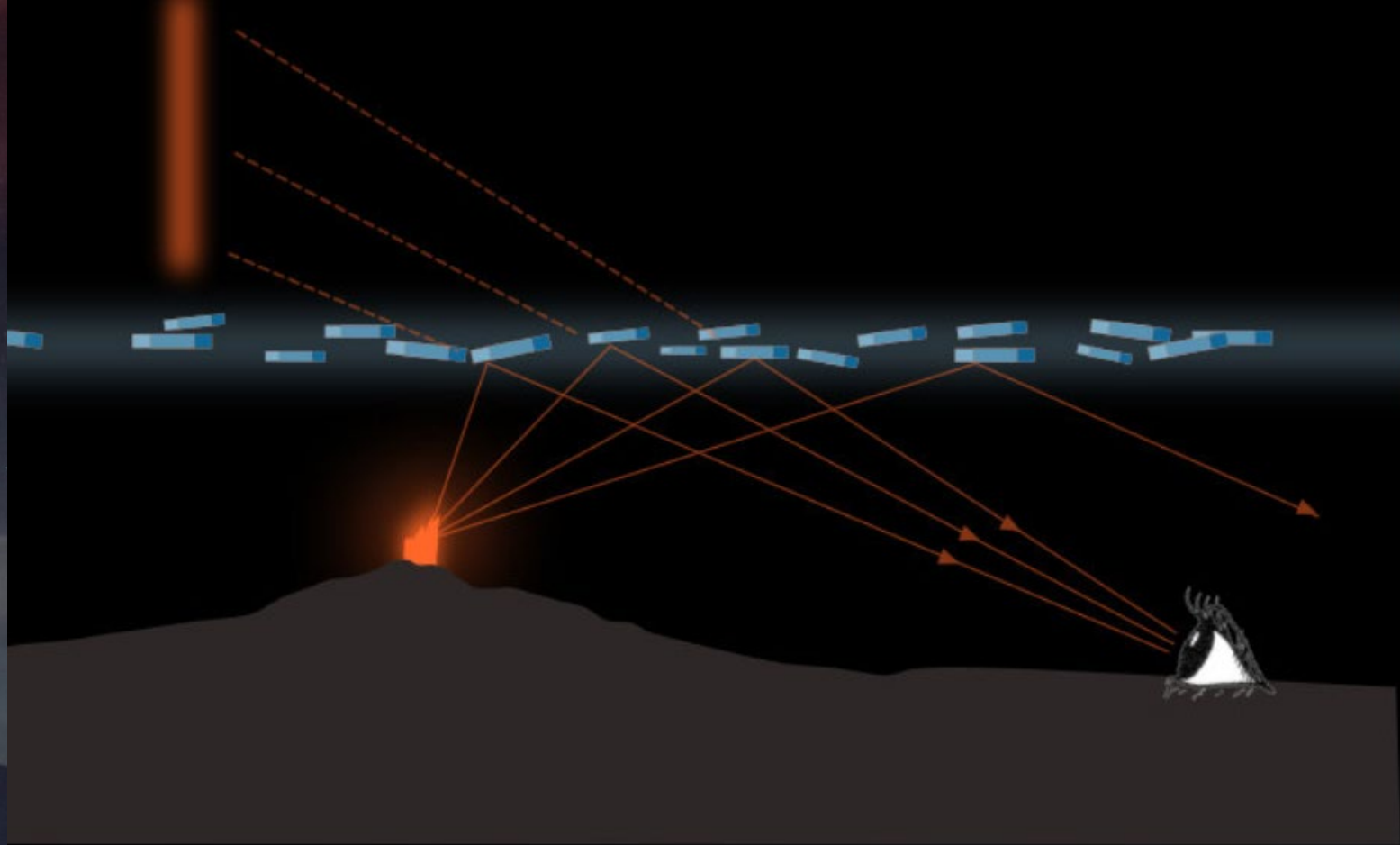


Stacked layers of aragonite tiles Regularly spaced surface grooves
The green bars represent 10 micron.

Images ©T L Tan, D Wong & P Lee, *Iridescence of a shell of mollusk Haliotis Glabra*, *Optics Express*, Vol. 12, Issue 20, pp. 4847-4854 (2004)



OPoD: Optics Picture of the Day | *Etna Light Pillar*



Q: 22.3

When laser light shines on a screen after passing through two closely spaced slits, you see

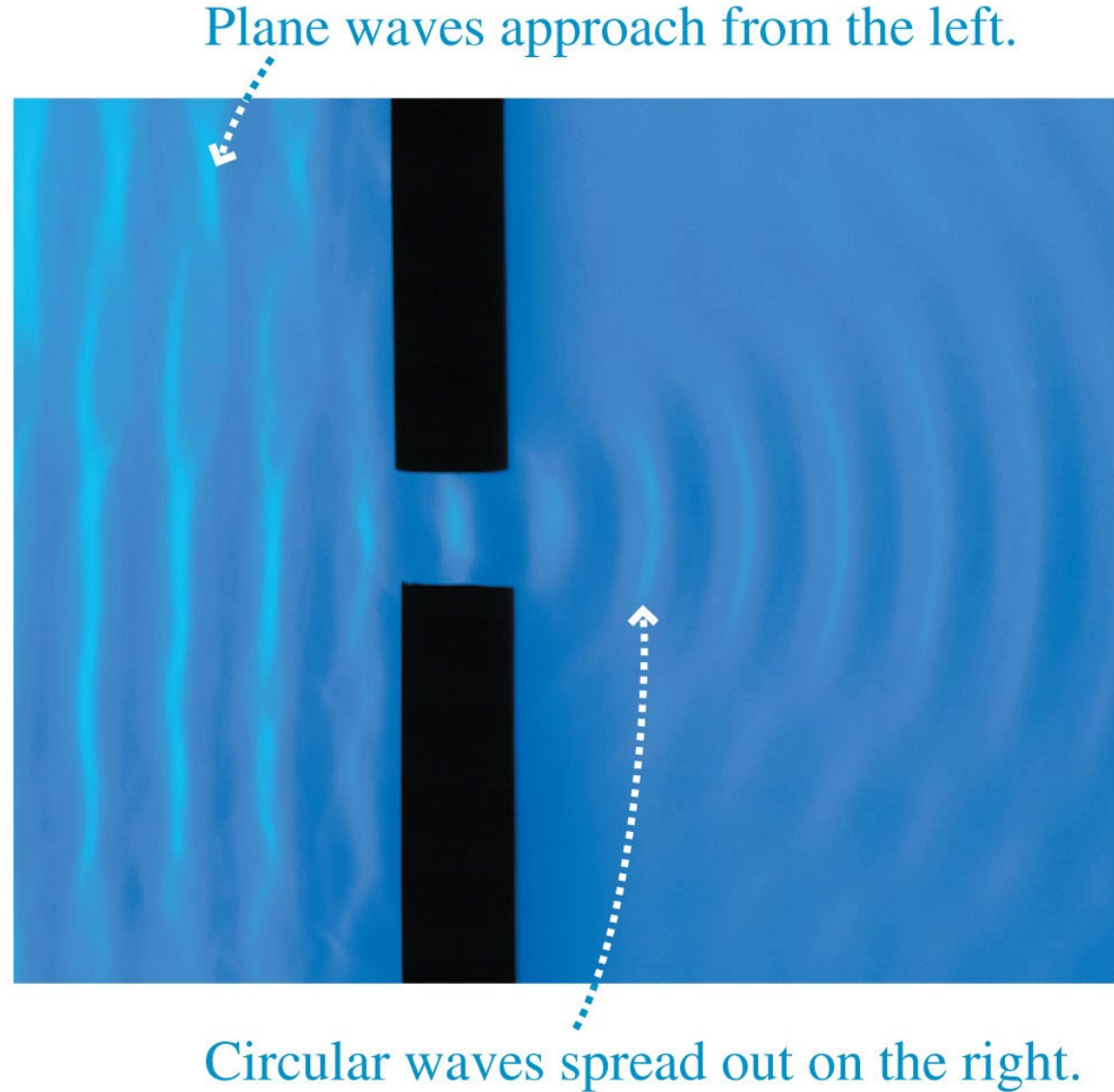
- A. An Airy disk.
- ✓ B. Interference fringes.
- C. Two dim, closely spaced points of light.
- D. A Poisson spot.
- E. An Arago spot.

The spreading of waves behind an aperture is

- ✓ A. More for long wavelengths, less for short wavelengths.
- B. Less for long wavelengths, more for short wavelengths.
- C. The same for long and short wavelengths.
- D. Non-existent.

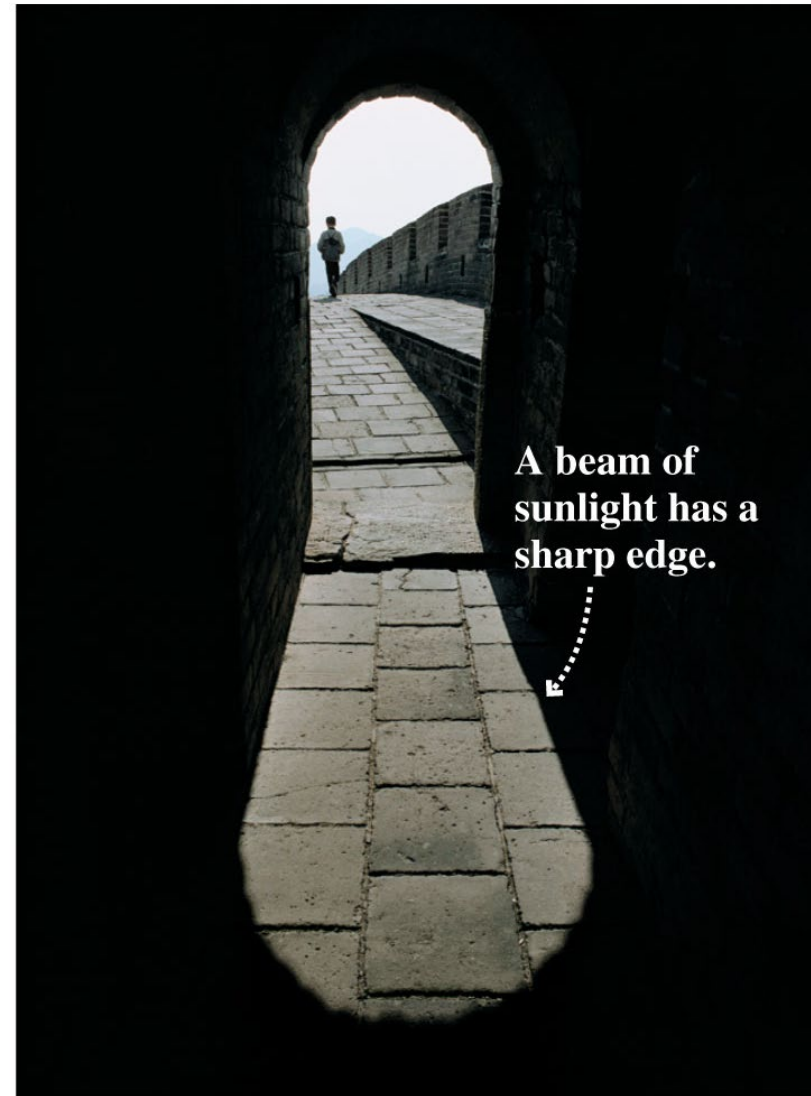
Diffraction of Water Waves

- A water wave, after passing through an opening, **spreads out** to fill the space behind the opening.
- This well-known spreading of waves is called **diffraction**.



Models of Light

- Unlike a water wave, when light passes through a large opening, it makes a sharp-edged shadow.
- This lack of noticeable diffraction means that if light is a wave, the wavelength must be very small.

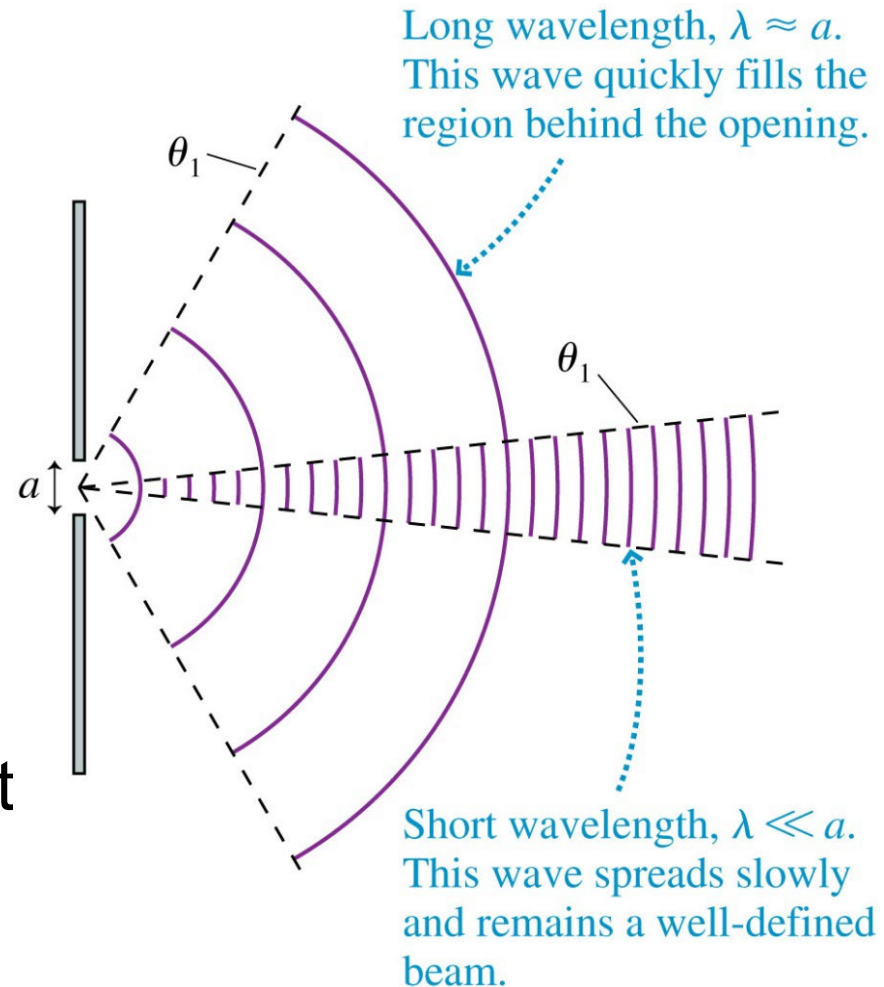


The Wave and Ray Models of Light

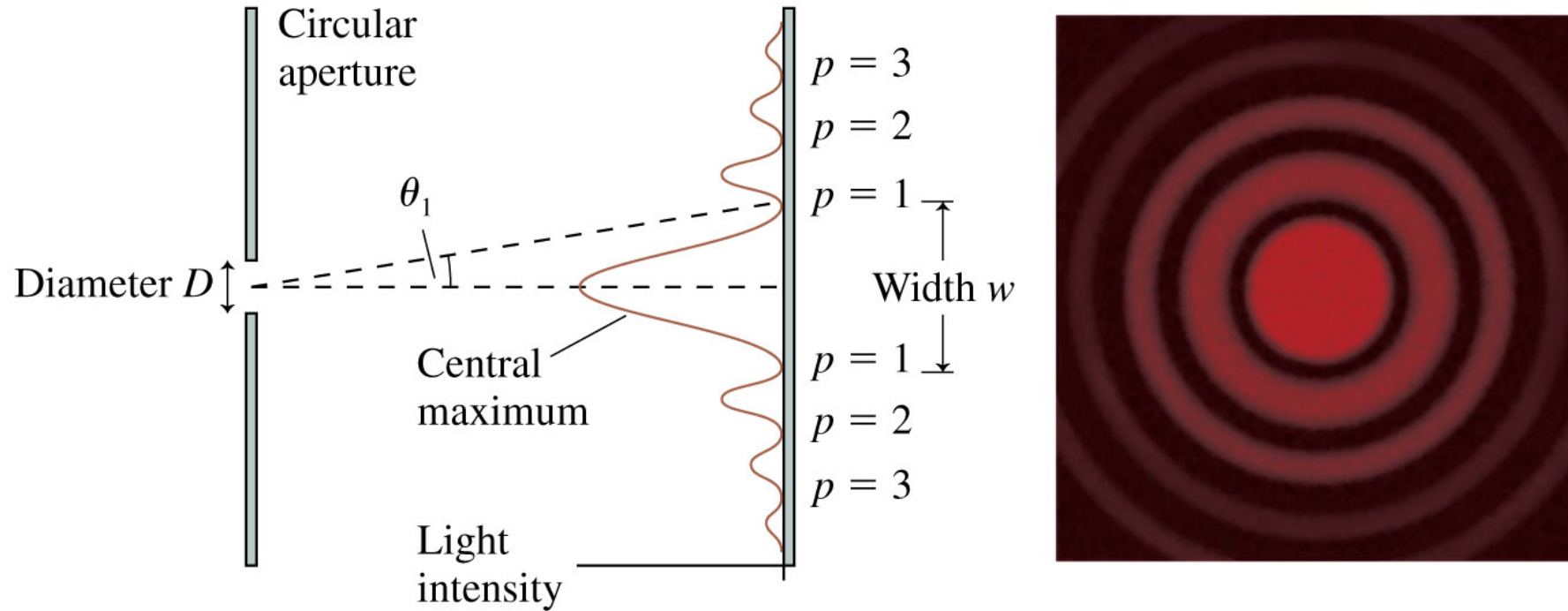
- When light passes through an opening of size a , the angle of the first diffraction minimum is:

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right)$$

- Light waves, because of their very short wavelength, almost always have $\lambda/a \ll 1$ and diffract to produce a slowly spreading “beam” of light.



The Diffraction of Light by a Circular Opening

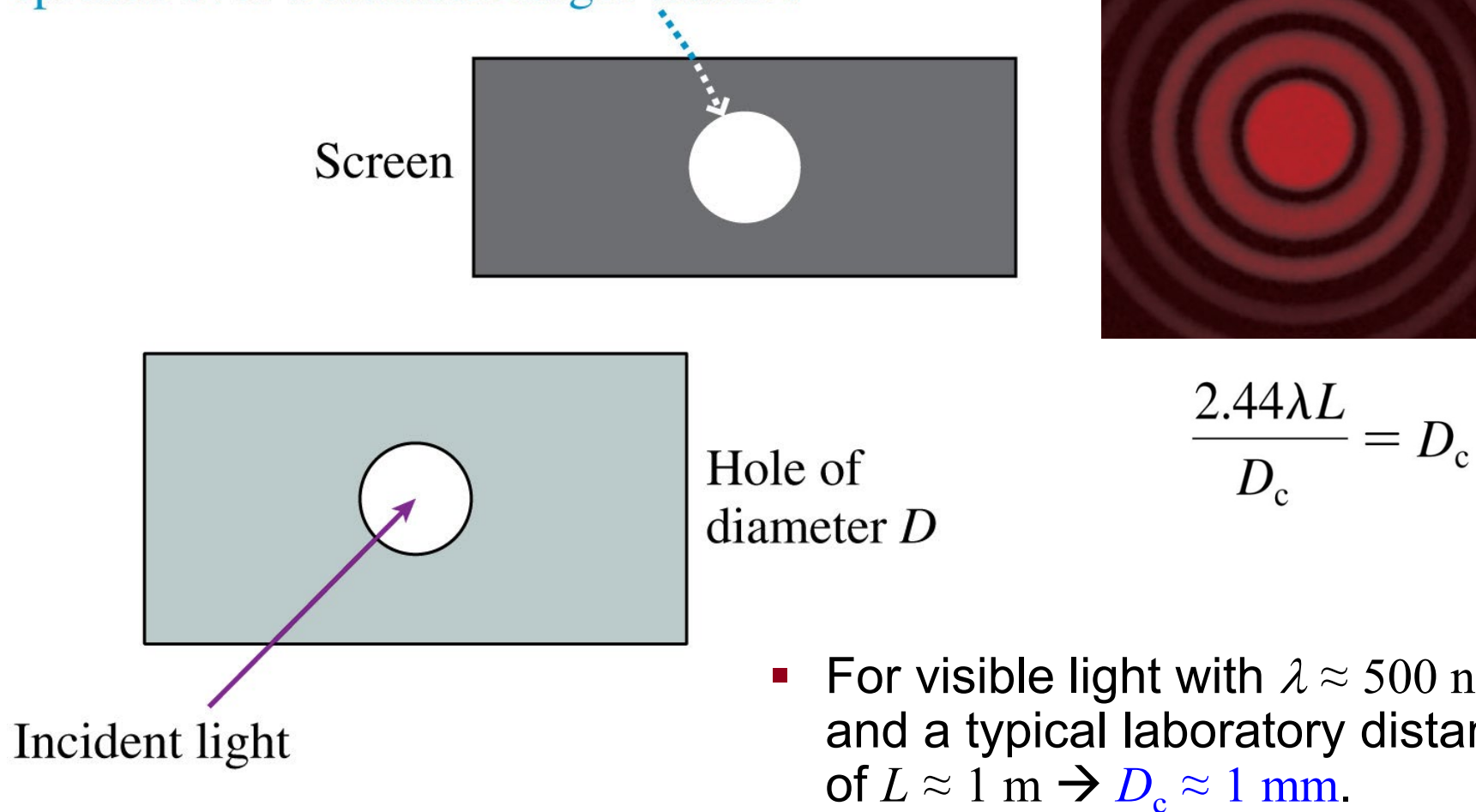


$$\theta_1 = \frac{1.22\lambda}{D}$$

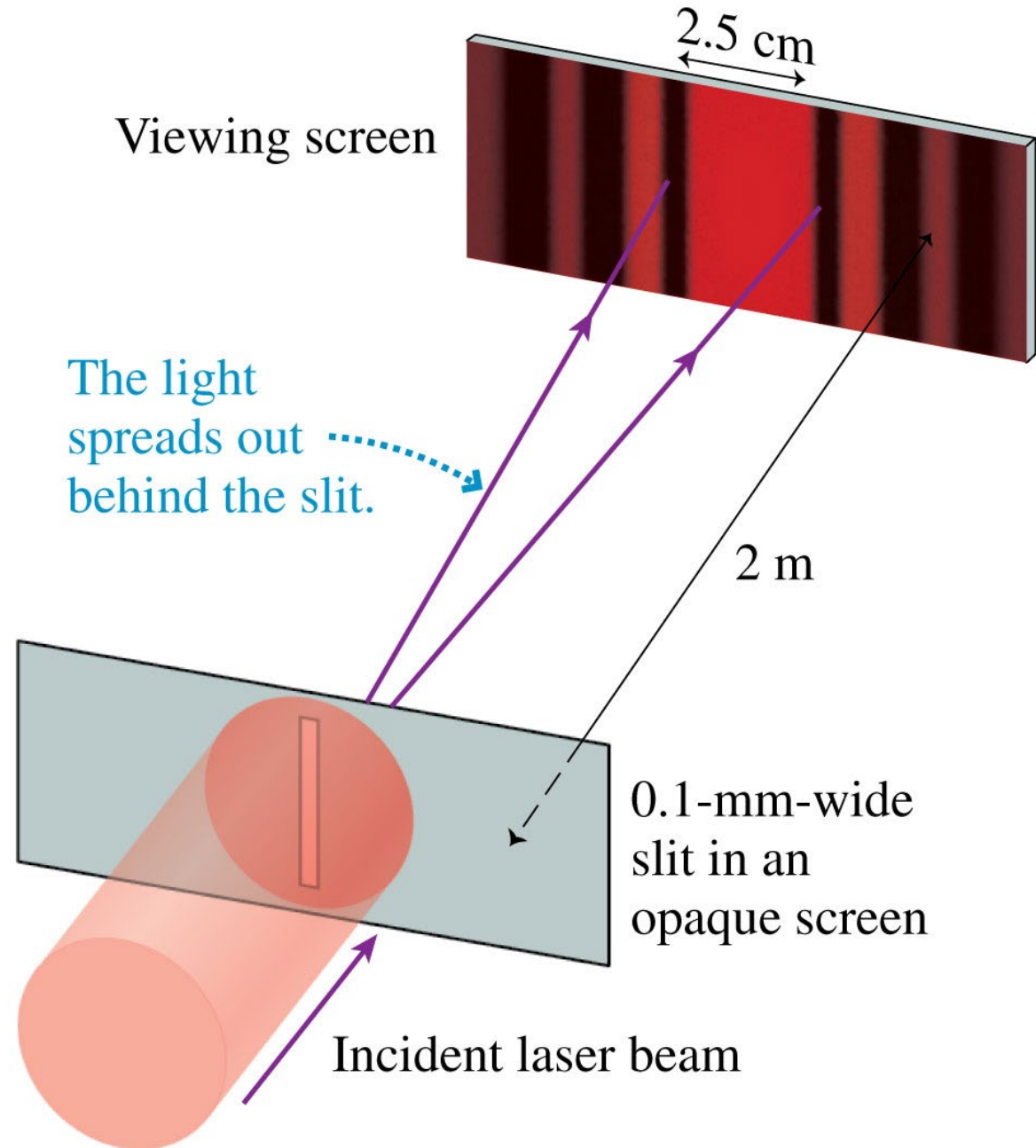
$$w = 2y_1 = 2L \tan \theta_1 \approx \frac{2.44\lambda L}{D}$$

The Wave and Ray Models of Light

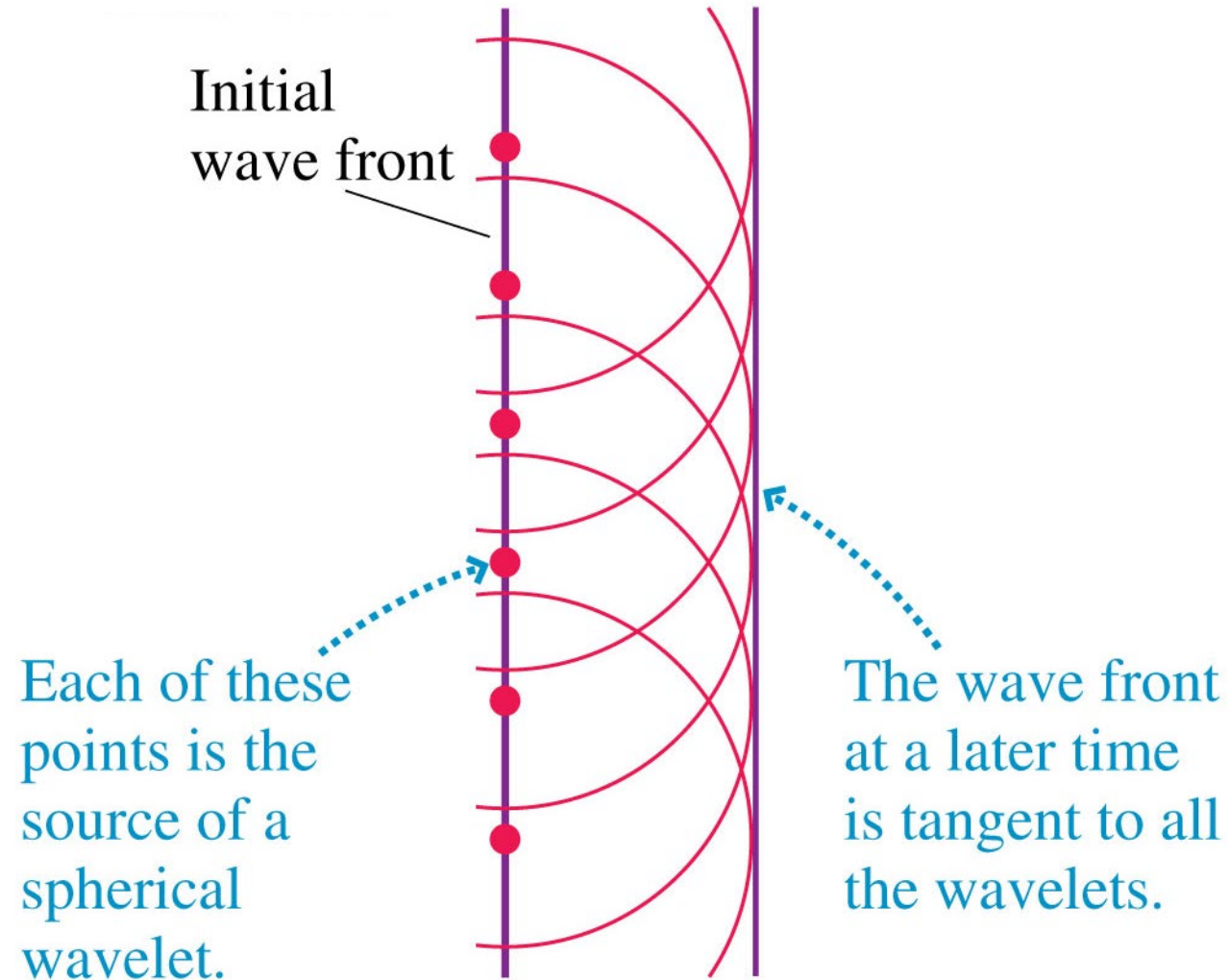
If light travels in straight lines, the image on the screen is the same size as the hole.
Diffraction will not be noticed unless the light spreads over a diameter larger than D .



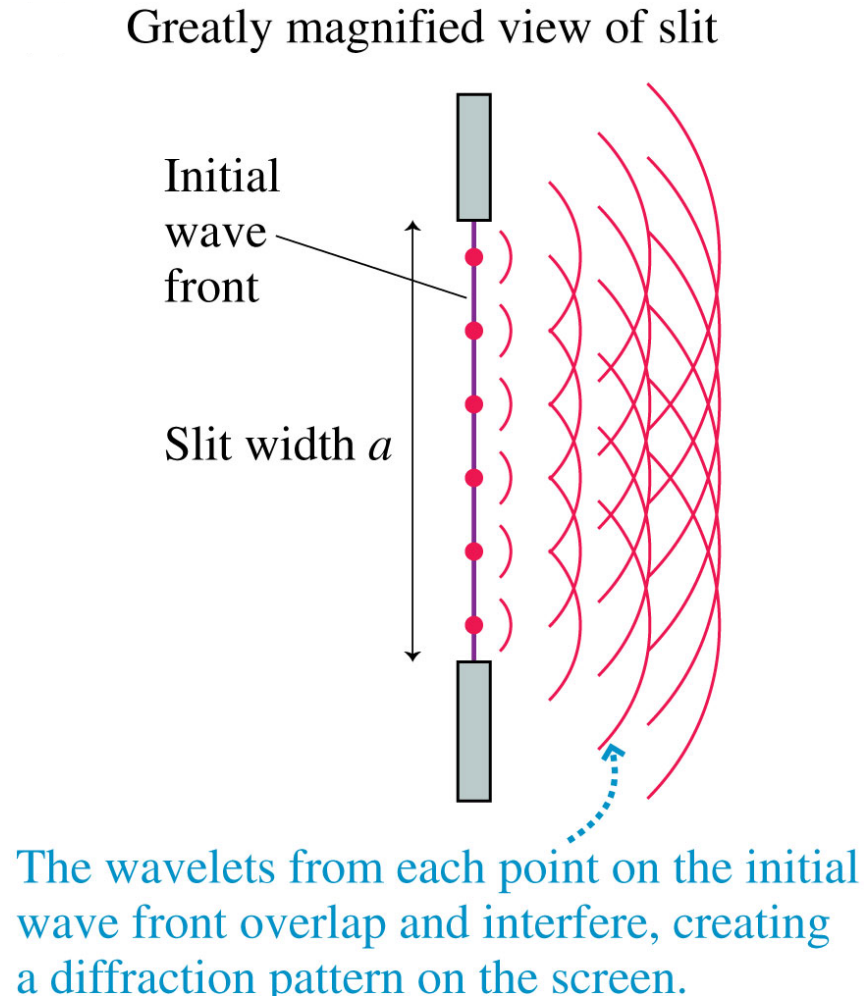
Diffraction of Light



Huygens' Principle: Plane Waves

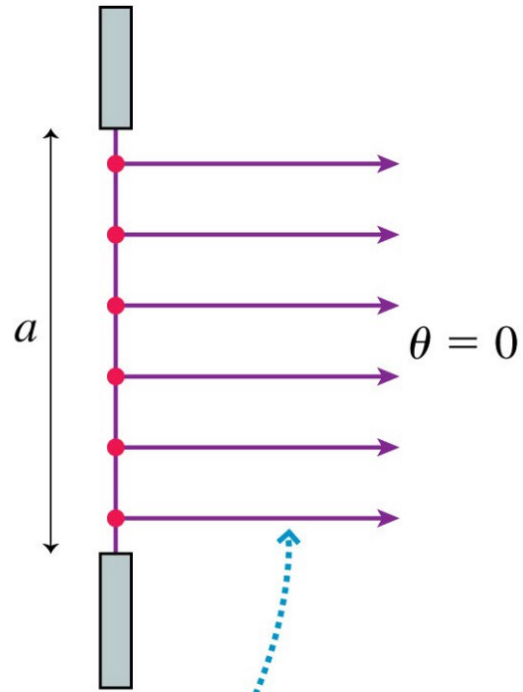


Analyzing Single-Slit Diffraction



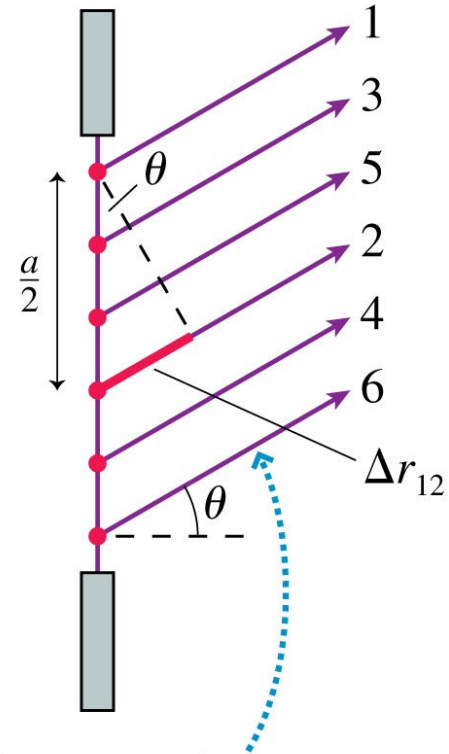
- The figure shows a wave front passing through a narrow slit of width a .
- According to Huygens' principle, each point on the wave front can be thought of as the source of a spherical wavelet.

Analyzing Single-Slit Diffraction



The wavelets going straight forward all travel the same distance to the screen. Thus they arrive in phase and interfere constructively to produce the central maximum.

Each point on the wave front is paired with another point distance $a/2$ away.



These wavelets all meet on the screen at angle θ . Wavelet 2 travels distance $\Delta r_{12} = (a/2) \sin \theta$ farther than wavelet 1.

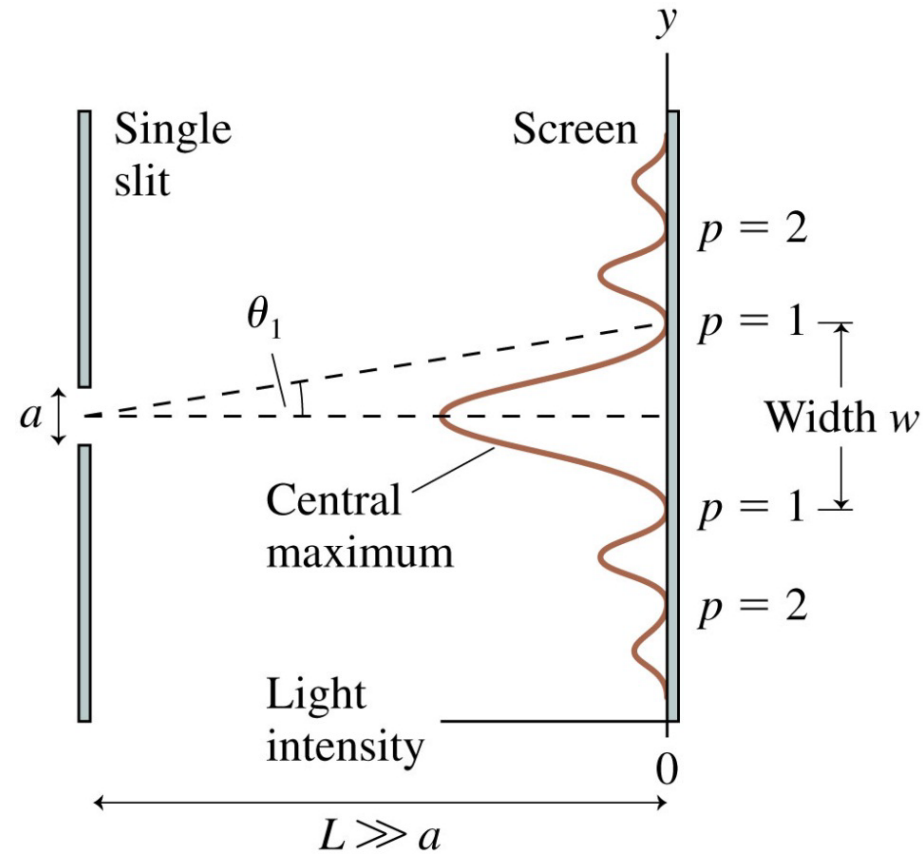
- If the **path-length difference** is $\Delta r = \lambda/2$, the wavelets arrive at the screen *out of phase* and interfere destructively.

Single-Slit Diffraction

- The light pattern from a single slit consists of a **central maximum** flanked by a series of weaker **secondary maxima** and dark fringes.
- The **dark fringes** occur for $a \sin \theta_p = p\lambda$, or ($\theta_p \ll 1$ rad):

$$\theta_p = p \frac{\lambda}{a} \quad p = 1, 2, 3, \dots \quad (\text{angles of dark fringes})$$

Minima between the bright fringes are at $y_p = \frac{p\lambda L}{a}$.



Q&A 22.9

A laboratory experiment produces a single-slit diffraction pattern on a screen. If the slit is made narrower, the bright fringes will be



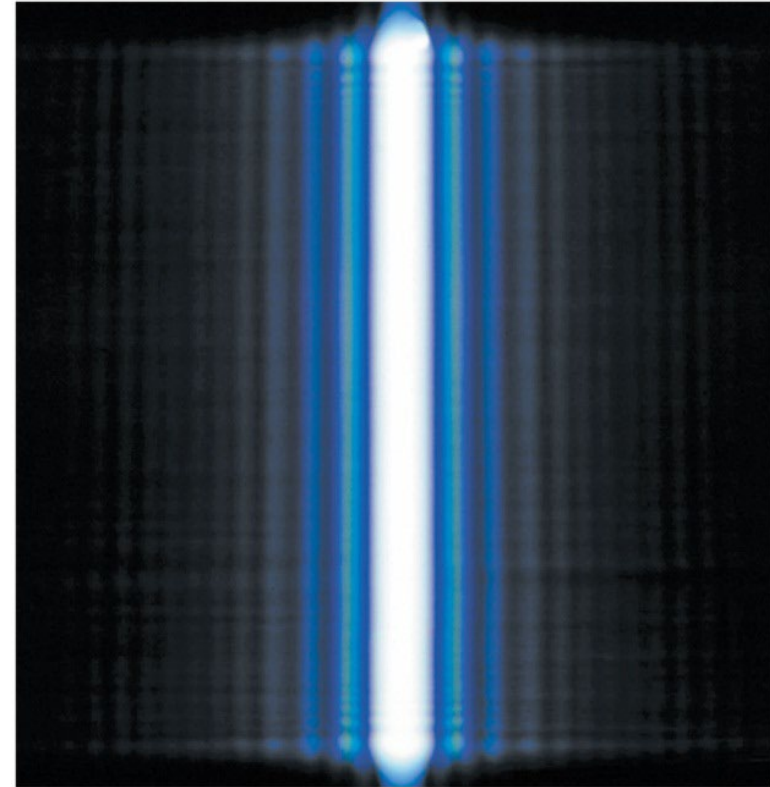
- ✓ A. Farther apart.
- B. In the same positions.
- C. Closer together.
- D. There will be no fringes because the conditions for diffraction won't be satisfied.

Minima between the bright fringes are at $y_p = \frac{p\lambda L}{a}$.

The Width of a Single-Slit Diffraction Pattern

- The central maximum of this single-slit diffraction pattern is much brighter than the secondary maximum.
- The width of the central maximum on a screen a distance L away is twice the spacing between the dark fringes on either side:

$$w = \frac{2\lambda L}{a} \quad (\text{single slit})$$



- The farther away from the screen (larger L), the wider the pattern of light becomes.
- The narrower the opening (smaller a), the wider the pattern of light becomes!

Q&A 22.11

A laboratory experiment produces a single-slit diffraction pattern on a screen. The slit width is a and the light wavelength is λ . In this case,



✓ A. $\lambda < a$.

B. $\lambda = a$.

C. $\lambda > a$.

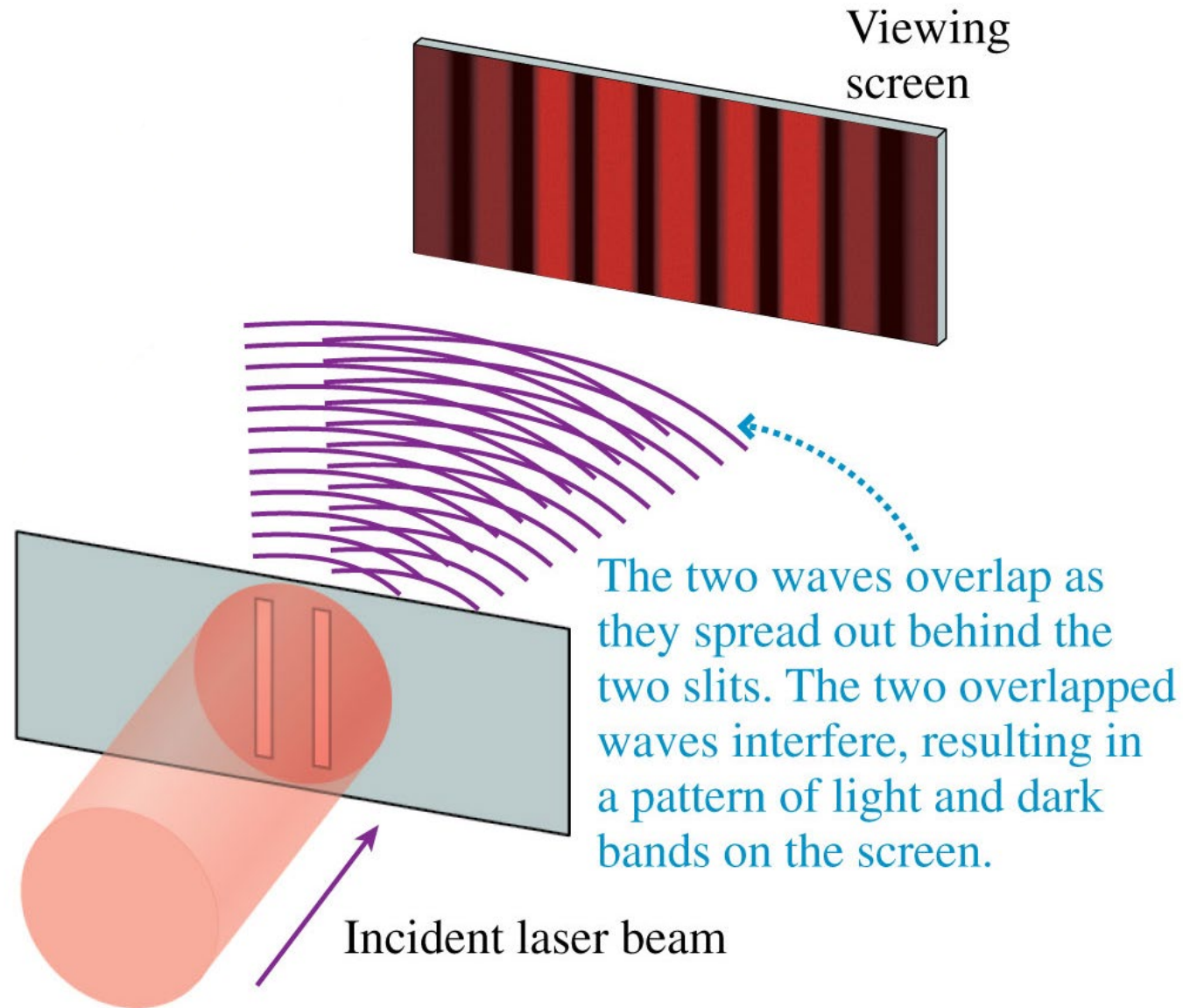
D. Not enough info to compare λ to a .

Dark fringes occur for $a \sin \theta_p = p\lambda$
 $\rightarrow \sin \theta_p < 1$, since $\theta_p < 90^\circ$

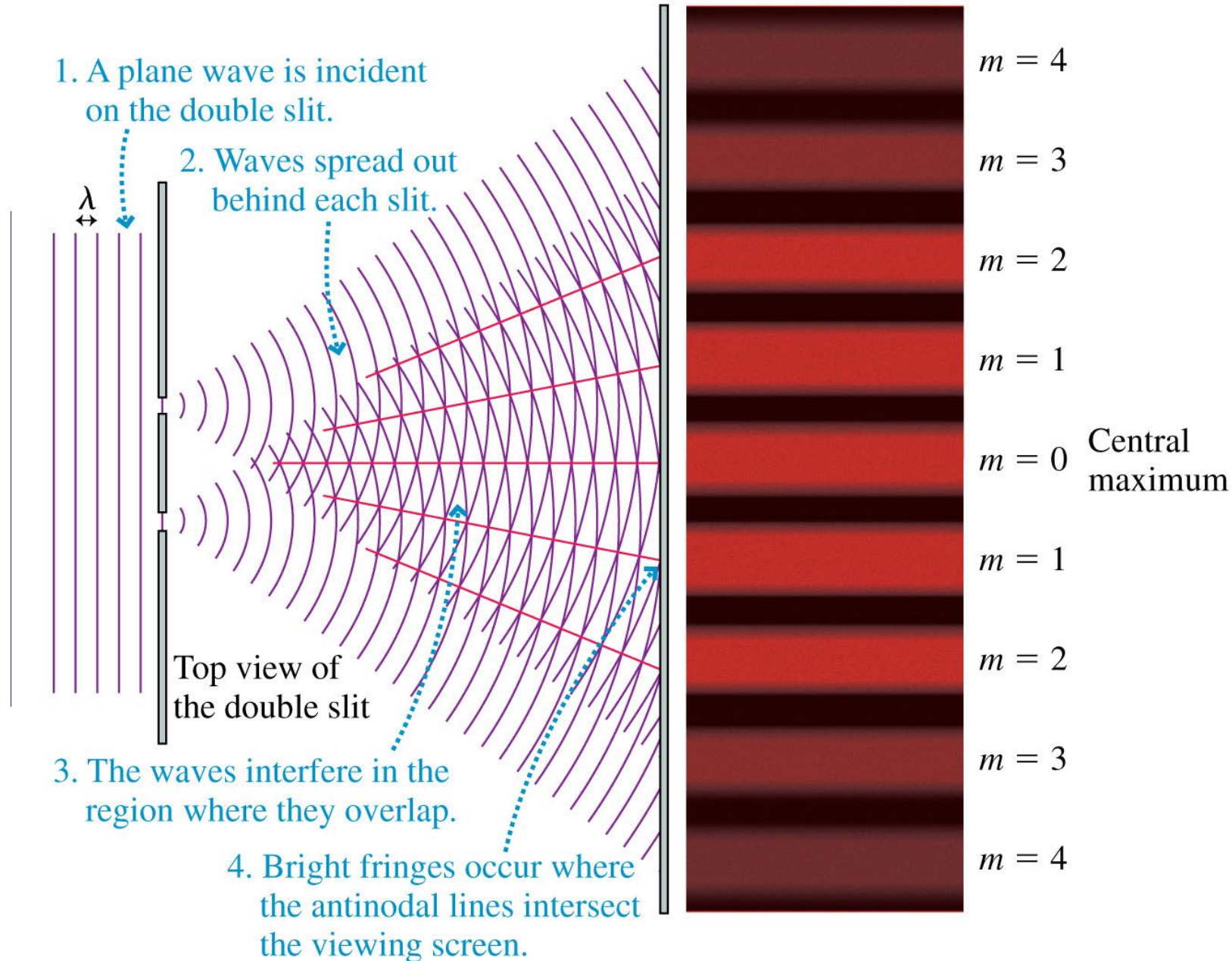
PhET: "Wave Interference"

The screenshot displays the PhET "Wave Interference" simulation interface. At the top, there are three tabs: "Water", "Sound", and "Light", with "Sound" currently selected. The main simulation area shows a speaker on the left emitting sound waves, represented by blue and black concentric circles. A vertical yellow barrier with two slits is positioned in the center. The waves are shown passing through the slits and creating an interference pattern on the right. On the left side of the speaker, there are controls for "On" (selected), "Off", and "Pulse". Below these is a "Wavelength" slider with a color spectrum bar and an "Amplitude" slider. At the bottom center, there is a "Show Graph" button and a play/pause button. On the right side, there is a "Screen" panel with various controls: "Measuring Tape" (checkbox), "Stopwatch" (checkbox), "Add Detector" (button), "Reset All" (button), "Rotate View" (slider between "Top" and "Side"), "One Light" (radio button, selected), "Two Lights" (radio button), "Spacing" (slider between 0 nm and 4200 nm), "No Barrier" (radio button), "One Slit" (radio button), "Two Slits" (radio button, selected), "Slit Width" (slider between 0 nm and 2100 nm), "Barrier Location" (slider between 0 nm and 5250 nm), "Slit Separation" (slider between 0 nm and 3500 nm), and "Add Mirror" (button). A small graph showing "Amplitude" vs "Time (s)" is also visible in the "Screen" panel.

Young's Double-Slit Experiment



Young's Double-Slit Experiment



Q: Sun Focused onto Retina

What is the intensity, in W/cm^2 , of the image of the Sun on the retina, as per Problem 2.11(a) in *Peatross & Ware*?

- A. 0.0014
- B. 0.014
- C. 0.14
- D. 1.4
- E. 14
- F. 140
- G. 1400
- H. 14000

Q: Laser Pointer Focused onto Retina

What is the intensity, in W/cm^2 , of the image of the 1 mW HeNe laser, as per Problem 2.11(b) in *Peatross & Ware*?

- A. 0.0039
- B. 0.039
- C. 0.39
- D. 3.9
- E. 39
- F. 390
- G. 3900
- H. 39000

Interference in Two and Three Dimensions

- The figure shows two identical sources that are in phase.
- The **path-length difference** Δr determines whether the interference at a particular point is constructive or destructive.

Maximum constructive interference:

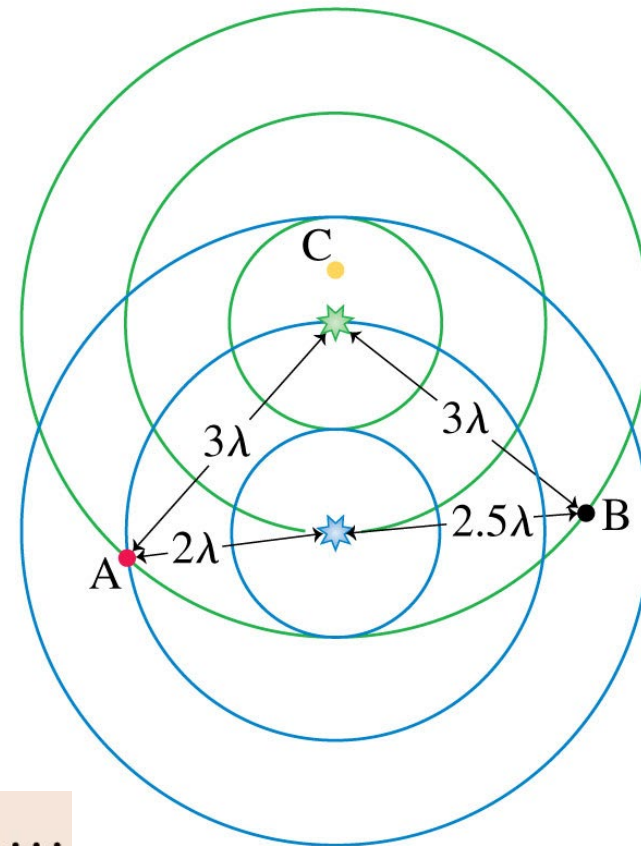
$$\Delta\phi = 2\pi \frac{\Delta r}{\lambda} + \Delta\phi_0 = m \cdot 2\pi$$

Perfect destructive interference:

$$\Delta\phi = 2\pi \frac{\Delta r}{\lambda} + \Delta\phi_0 = \left(m + \frac{1}{2}\right) \cdot 2\pi$$

$$m = 0, 1, 2, \dots$$

- At A, $\Delta r_A = \lambda$, so this is a point of constructive interference.



- At B, $\Delta r_B = \frac{1}{2}\lambda$, so this is a point of destructive interference.

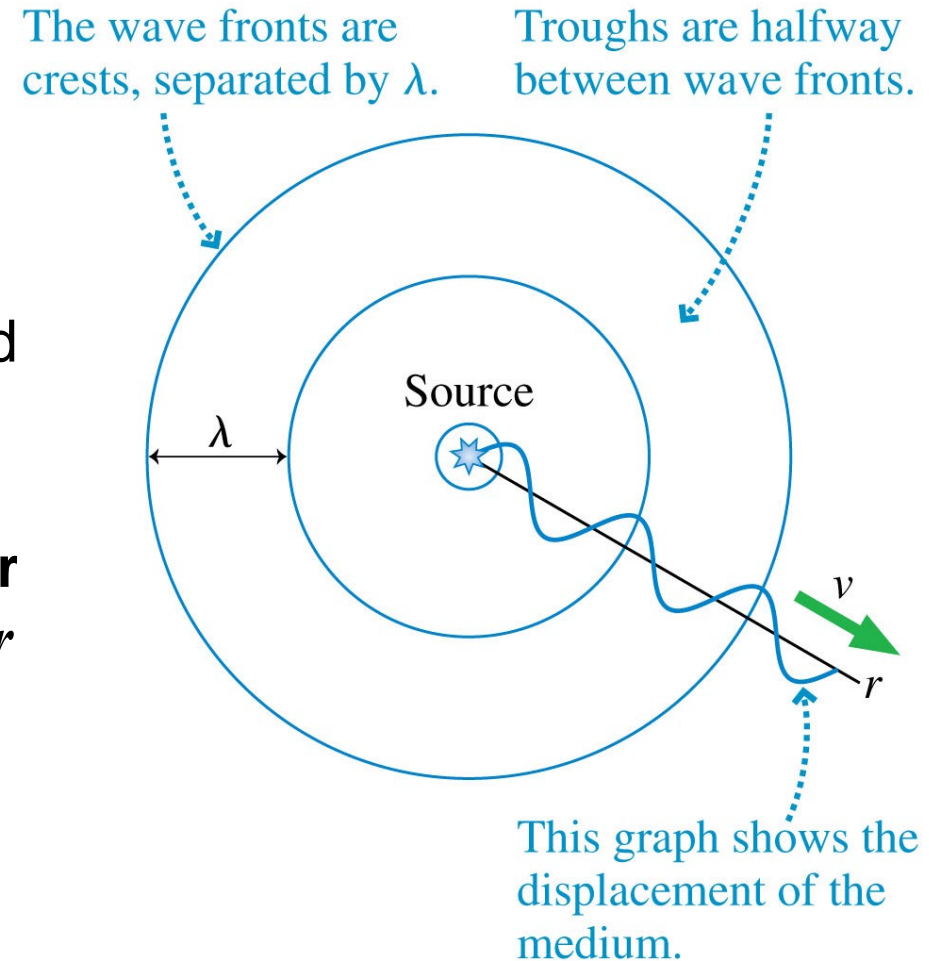
A Circular or Spherical Wave

- A circular or spherical wave can be written:

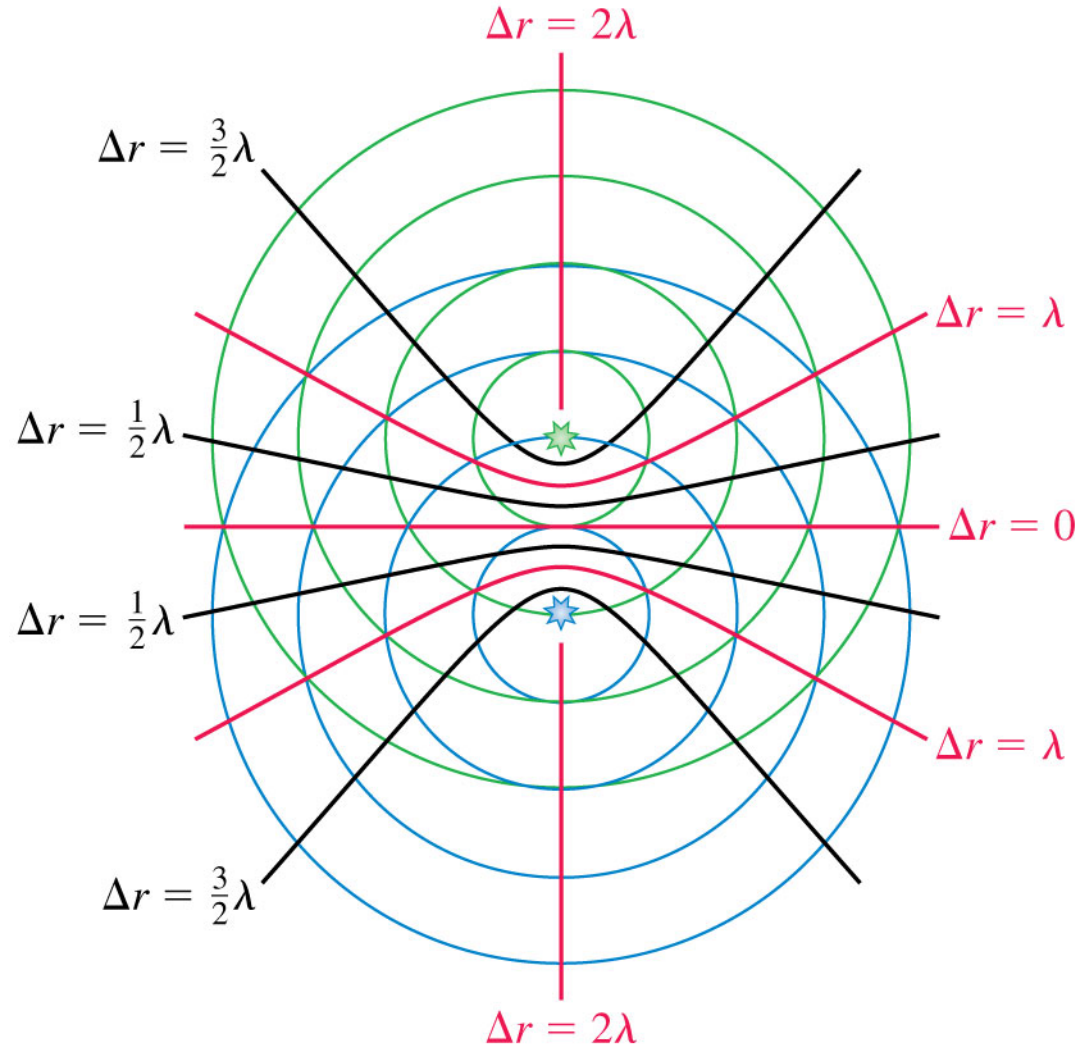
$$D(r, t) = a(r) \sin(kr - \omega t + \phi_0)$$

where r is the distance measured outward from the source.

- **The amplitude a of a circular or spherical wave diminishes as r increases.**



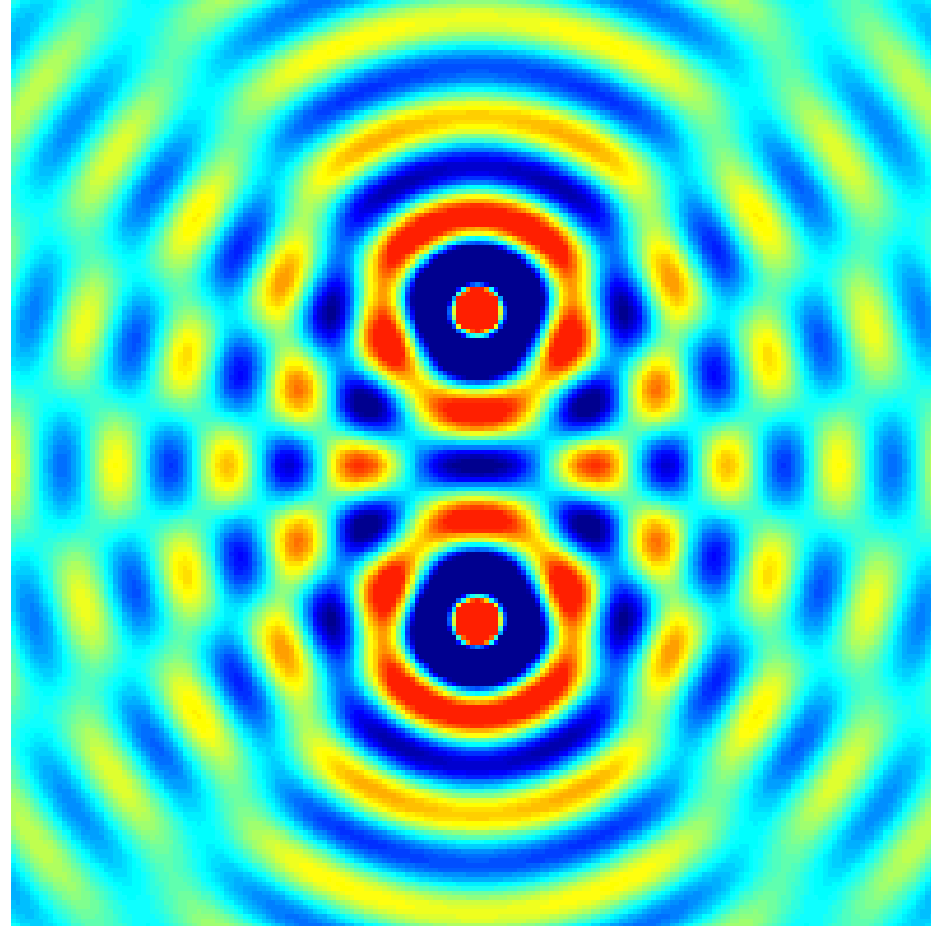
Interference in Two and Three Dimensions



- Antinodal lines, constructive interference, oscillation with maximum amplitude. Intensity is at its maximum value.
- Nodal lines, destructive interference, no oscillation. Intensity is zero.



Picturing Interference: Two identical sources

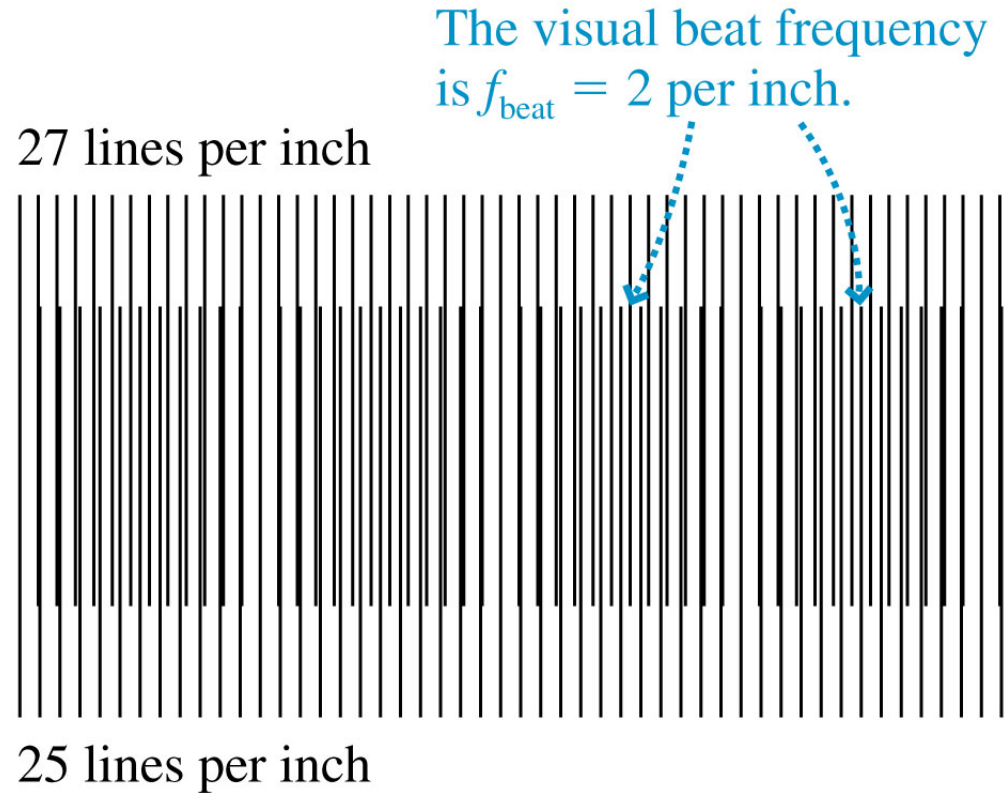


From [http://en.wikipedia.org/wiki/Interference_\(wave_propagation\)](http://en.wikipedia.org/wiki/Interference_(wave_propagation))

Visual Beats

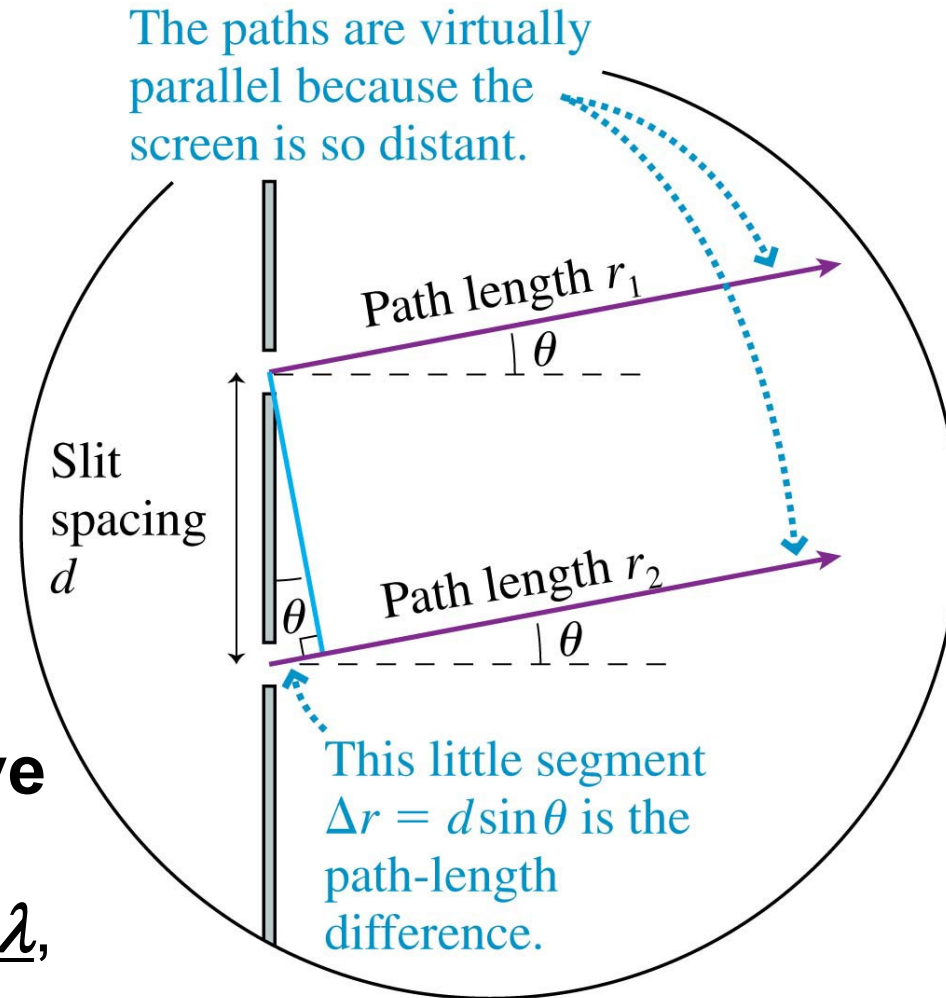
- Shown is a graphical example of beats.
- Two “fences” of slightly different frequencies are superimposed on each other.
- The center part of the figure has two “beats” per inch:

$$f_{\text{beat}} = 27 - 25 = 2$$



Analyzing Double-Slit Interference

- The figure shows a magnified portion of the double-slit experiment.
- The wave from the lower slit travels an extra distance:
$$\Delta r = d \sin \theta$$
- Bright fringes (**constructive interference**) will occur at angles θ_m such that $\Delta r = m\lambda$, where $m = 0, 1, 2, 3, \dots$



Analyzing Double-Slit Interference

- The m th bright fringe emerging from the double slit is at an angle:

$$\theta_m = m \frac{\lambda}{d} \quad m = 0, 1, 2, 3, \dots \quad (\text{angles of bright fringes})$$

where θ_m is in radians, and we have used the **small-angle approximation**.

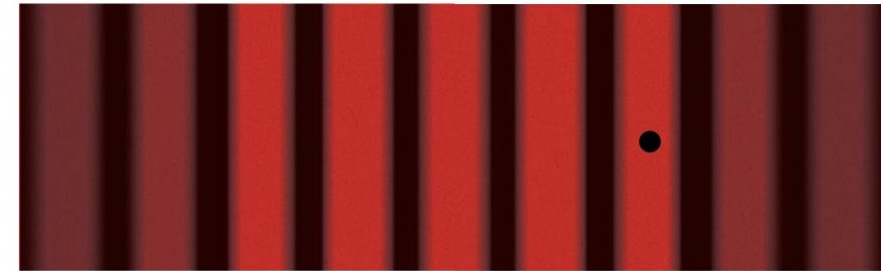
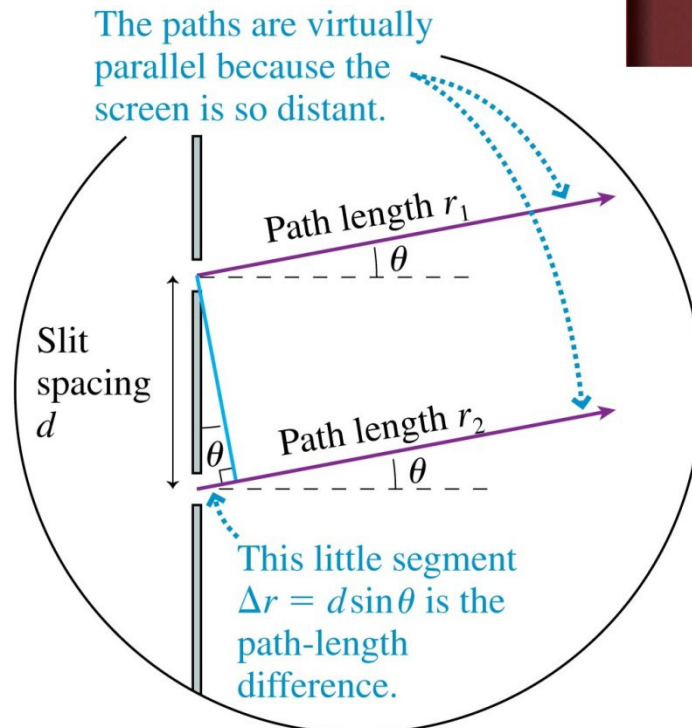
- The y -position on the screen of the m th bright fringe on a screen a distance L away is:

$$y_m = \frac{m\lambda L}{d} \quad m = 0, 1, 2, 3, \dots \quad (\text{positions of bright fringes})$$

Q&A 22.3

A laboratory experiment produces a double-slit interference pattern on a screen. The point on the screen marked with a dot is how much farther from the left slit than from the right slit?

- A. 1.0λ .
- B. 1.5λ .
- ✓ C. 2.0λ .
- D. 2.5λ .
- E. 3.0λ .



Central maximum

- The wave from the lower slit travels an extra distance:

$$\Delta r = d \sin \theta$$

- Bright fringes (constructive interference) will occur at angles θ_m such that $\Delta r = m\lambda$, where $m = 0, 1, 2, 3, \dots$

Q&A 22.4

A laboratory experiment produces a double-slit interference pattern on a screen. If the screen is moved farther away from the slits, the fringes will be:

- A. Closer together.
- B. In the same positions.
- ✓ C. Farther apart.
- D. Fuzzy and out of focus.



Central maximum

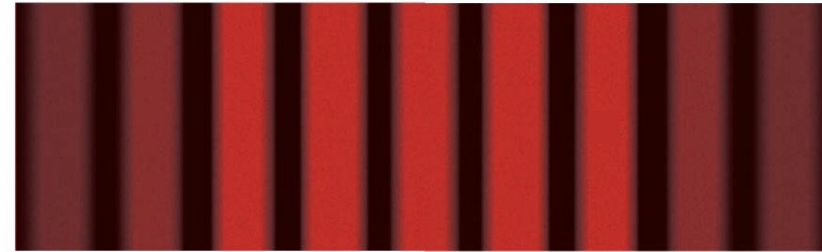
$$\Delta y = \frac{\lambda L}{d}$$

$$y_m = \frac{m\lambda L}{d} \quad m = 0, 1, 2, 3, \dots \quad (\text{positions of bright fringes})$$

Q&A 22.5

A laboratory experiment produces a double-slit interference pattern on a screen. If **green light** is used, with everything else the same, the bright fringes will be

- ✓ A. Closer together.
- B. In the same positions.
- C. Farther apart.
- D. There will be no fringes because the conditions for interference won't be satisfied.

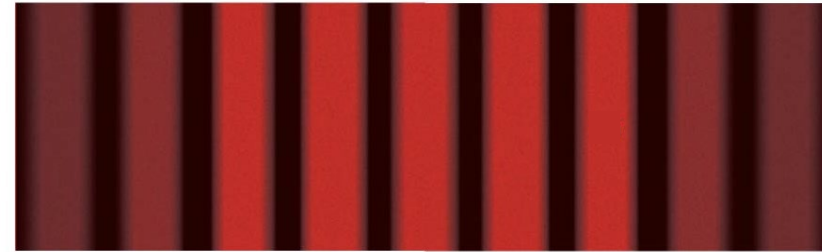


$$\Delta y = \frac{\lambda L}{d} \text{ and green light has a shorter wavelength.}$$

Q&A 22.6

A laboratory experiment produces a double-slit interference pattern on a screen. If the slits are moved closer together, the bright fringes will be:

- A. Closer together.
- B. In the same positions.
- ✓ C. Farther apart.
- D. There will be no fringes because the conditions for interference won't be satisfied.



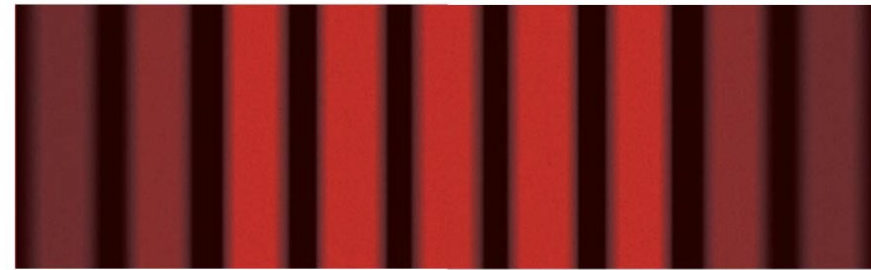
$$\Delta y = \frac{\lambda L}{d} \text{ and } d \text{ is smaller.}$$

Q&A 22.7

A laboratory experiment produces a double-slit interference pattern on a screen. If the amplitude of the light wave is doubled, the intensity of the central maximum will increase by a factor of:

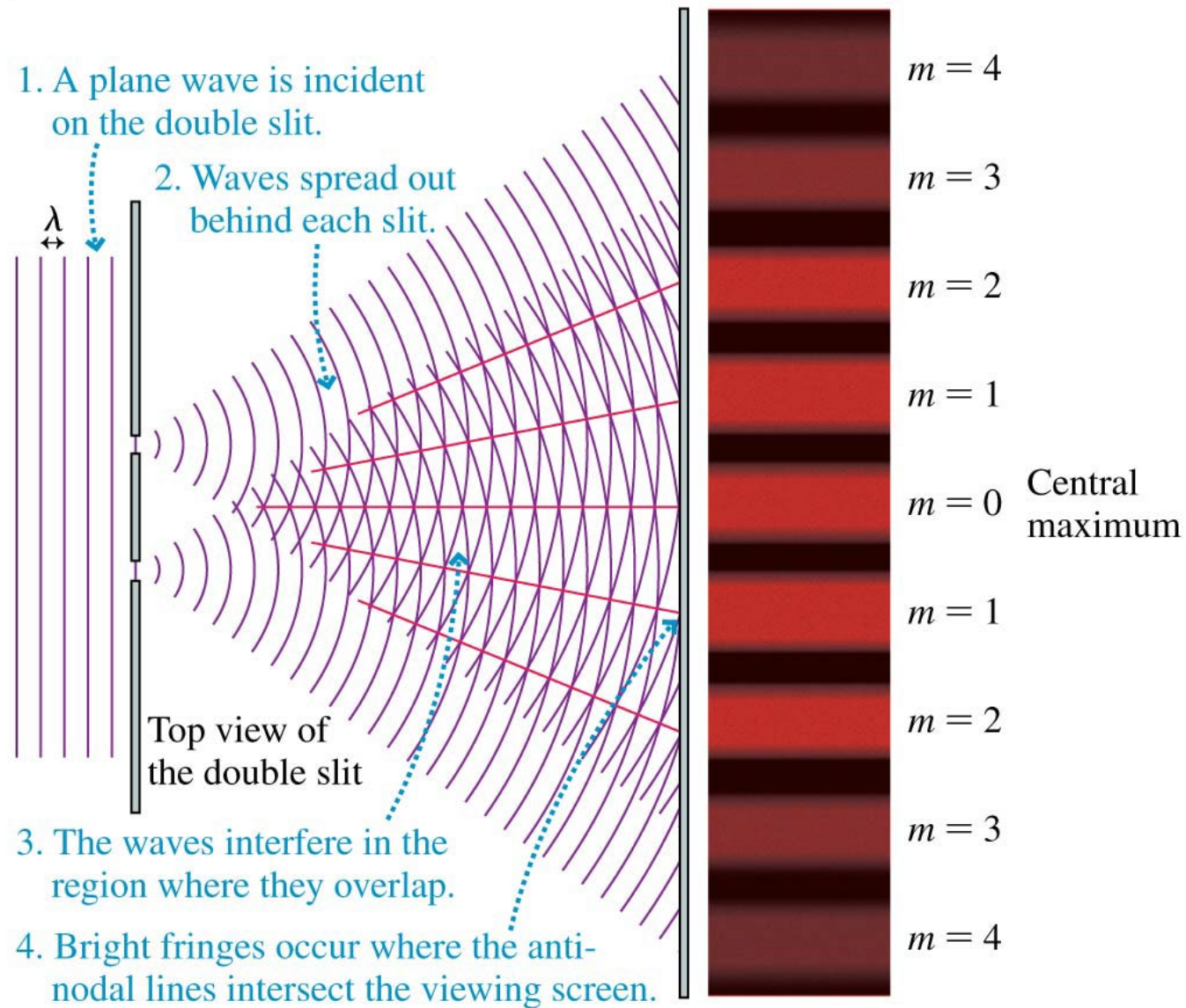
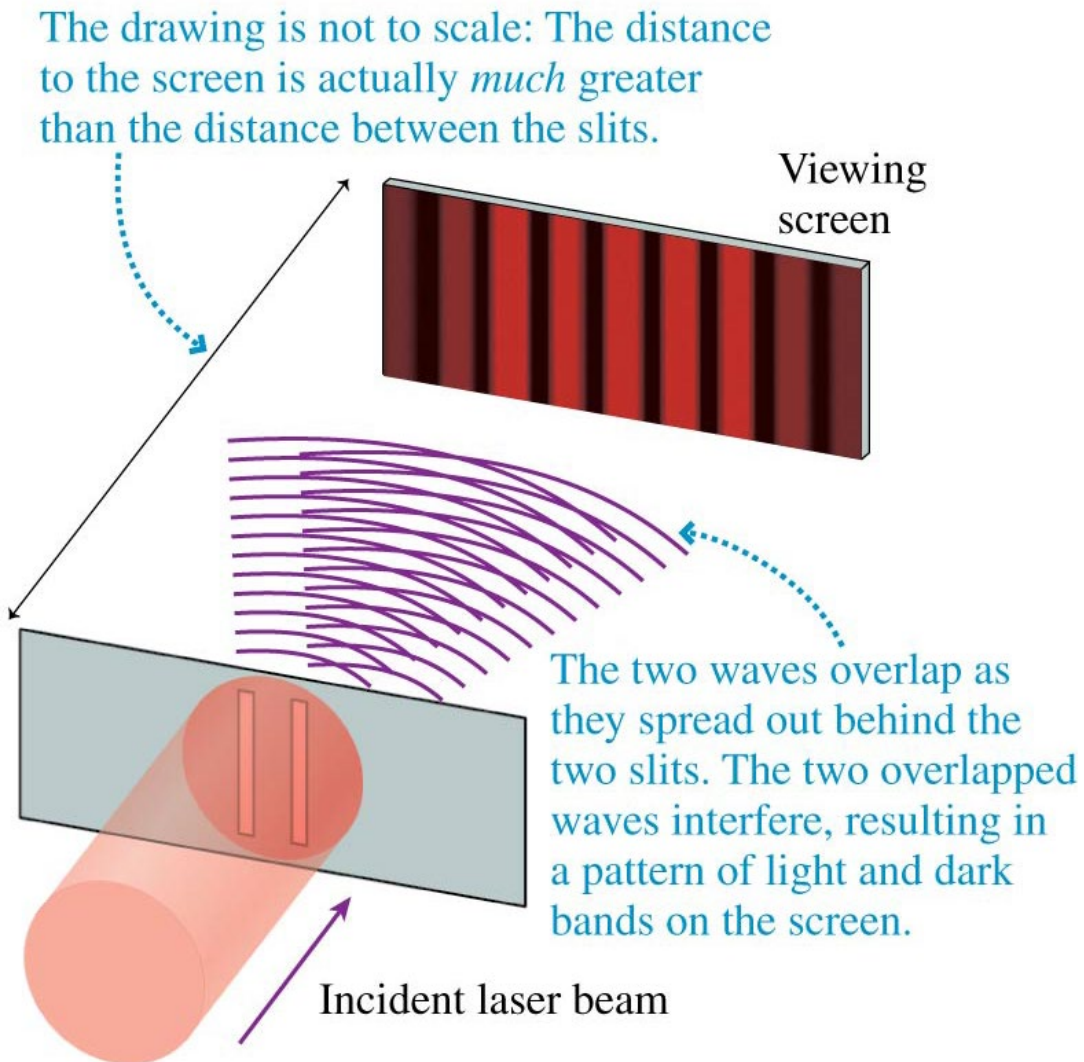
- A. $\sqrt{2}$.
- B. 2.
- ✓ C. 4.
- D. 8.

$$I \propto A^2$$



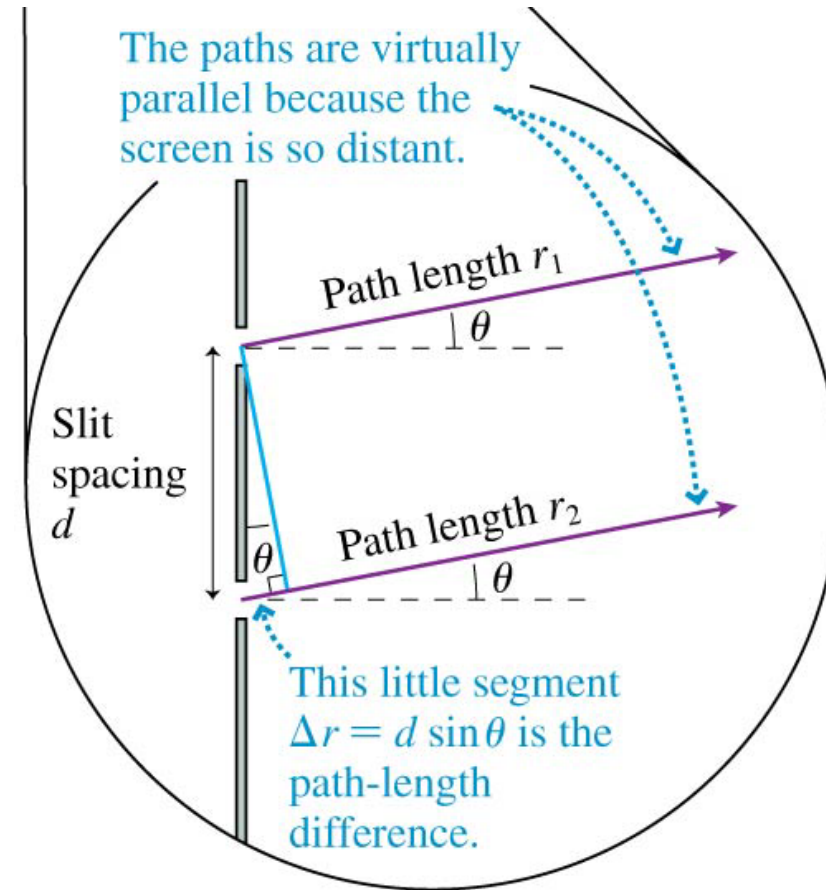
Central maximum

Young's Double-Slit Experiment... Again



Analyzing Double-Slit Interference

- The figure shows a magnified portion of the double-slit experiment.
- The wave from the lower slit travels an extra distance:
$$\Delta r = d \sin \theta$$
- Bright fringes (constructive interference) will occur at angles θ_m such that $\Delta r = m\lambda$, where $m = 0, 1, 2, 3, \dots$



Intensity of the Double-Slit Interference Pattern

09/26/22

Intensity of Double-Slit Interference Pattern

Knight
4th ed.
p. 936

$$E = E_1 + E_2 = e \sin \phi_1 + e \sin \phi_2$$

$$\rightarrow kx_1 - \omega t + \phi_{10} \quad kx_2 - \omega t + \phi_{20}$$

Trig: $\sin \alpha + \sin \beta \equiv 2 \cos \left[\frac{1}{2}(\alpha - \beta) \right] \sin \left[\frac{1}{2}(\alpha + \beta) \right]$

1. $\phi_1 \neq \phi_2 \neq \phi_3$

$$\Delta \phi = \phi_2 - \phi_1 \rightarrow (\phi_0)_{\text{avg}} = \frac{1}{2}(\phi_{10} + \phi_{20})$$

$$\therefore E = \left[2e \cos \left(\frac{\Delta \phi}{2} \right) \right] \sin \left(kx_{\text{avg}} - \omega t + (\phi_0)_{\text{avg}} \right)$$

$$x_{\text{avg}} = \frac{1}{2}(x_1 + x_2)$$

$$A = | \dots |$$

$$\Delta \phi = m \cdot 2\pi \Rightarrow A = 2e \Rightarrow \text{in phase}$$

$$\Delta \phi = \left(m + \frac{1}{2}\right) \cdot 2\pi \Rightarrow A = 0 \Rightarrow \text{out of phase}$$

$$\Delta \phi = 90^\circ \Rightarrow A = \sqrt{2}e \Rightarrow \text{in between}$$

$$\Delta \phi = 2\pi \frac{\Delta x}{\lambda} + \Delta \phi_0$$

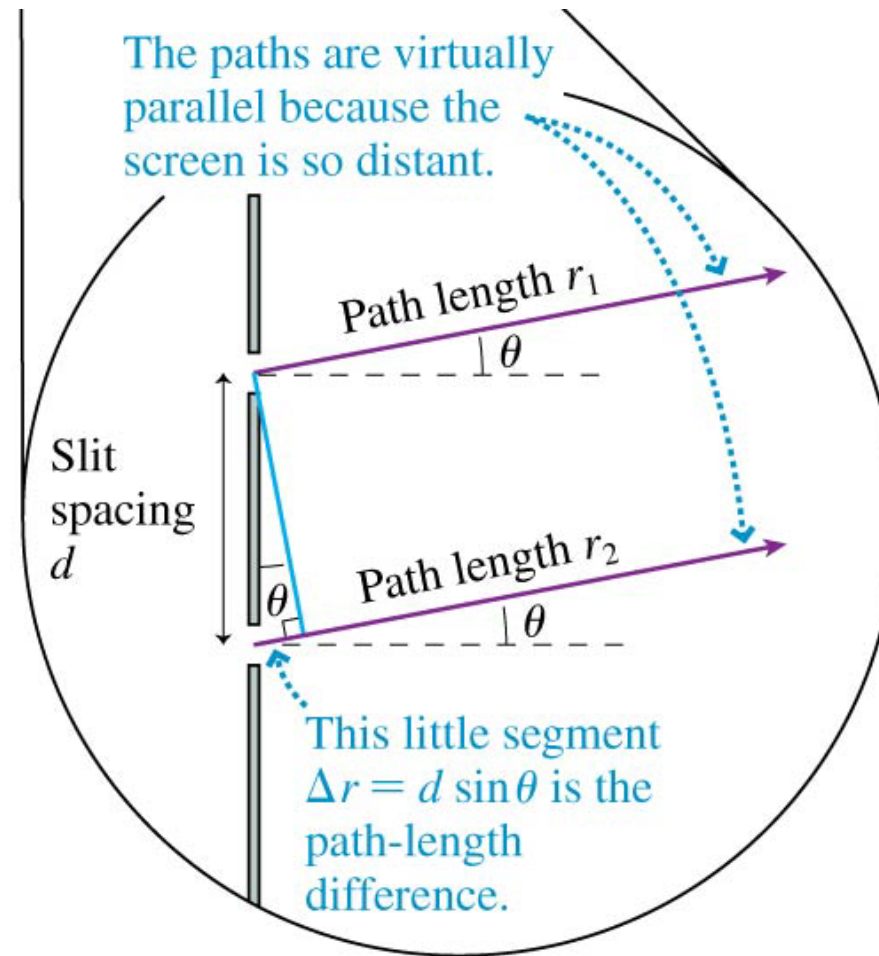
Double slit:
(ideal)

constant over slits $\Delta \phi_0 = 0 \Rightarrow$ slits in phase

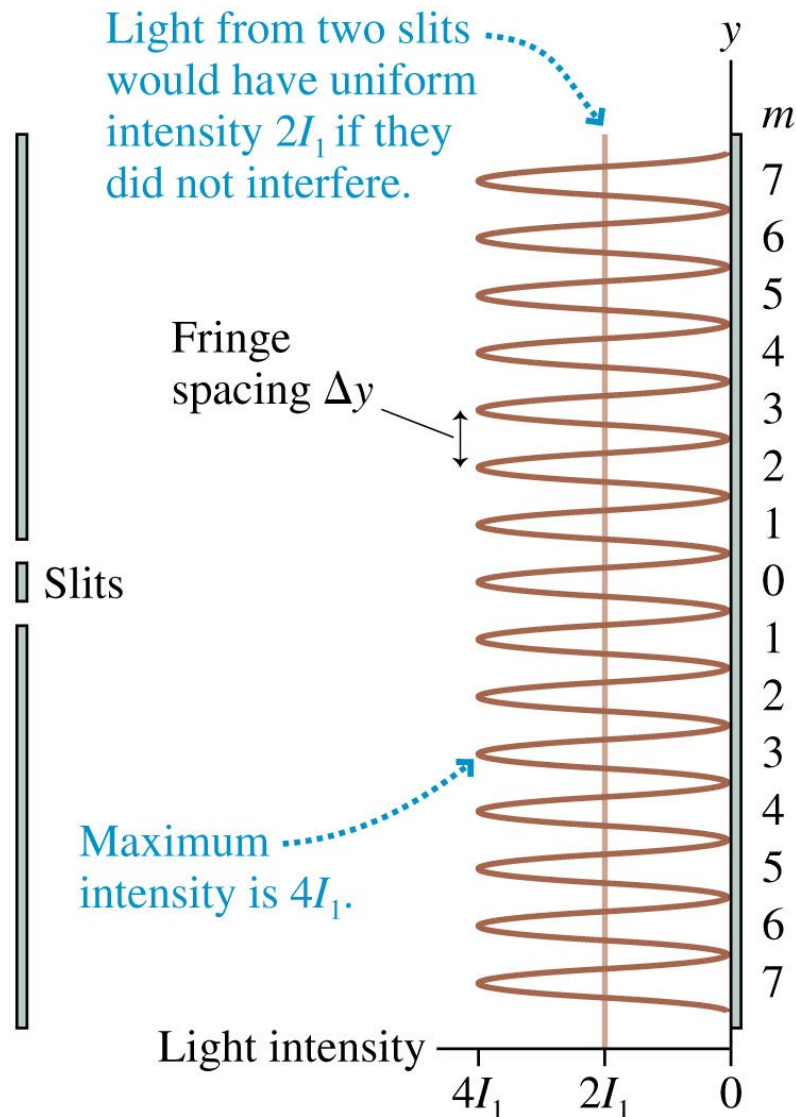
$$E = \left| 2e \cos \left(\frac{\Delta \phi}{2} \right) \right|, \text{ where } \Delta \phi = 2\pi \left(\frac{\Delta r}{\lambda} \right) \text{ and } \Delta r = d \sin \theta \Rightarrow$$

$$\Rightarrow \Delta \phi = 2\pi \frac{d \sin \theta}{\lambda} \approx 2\pi \frac{d \tan \theta}{\lambda} = 2\pi \frac{d (y/L)}{\lambda} = \frac{2\pi d}{\lambda} \cdot \frac{y}{L}$$

$$I = C E^2 = C^2 \left| 2e \cos \left(\frac{\pi d}{\lambda} \frac{y}{L} \right) \right|^2 = \underbrace{4 I_1}_{C e^2} \cos^2 \left(\frac{\pi d}{\lambda} \frac{y}{L} \right) \quad \text{N.B. No diffraction envelope.}$$



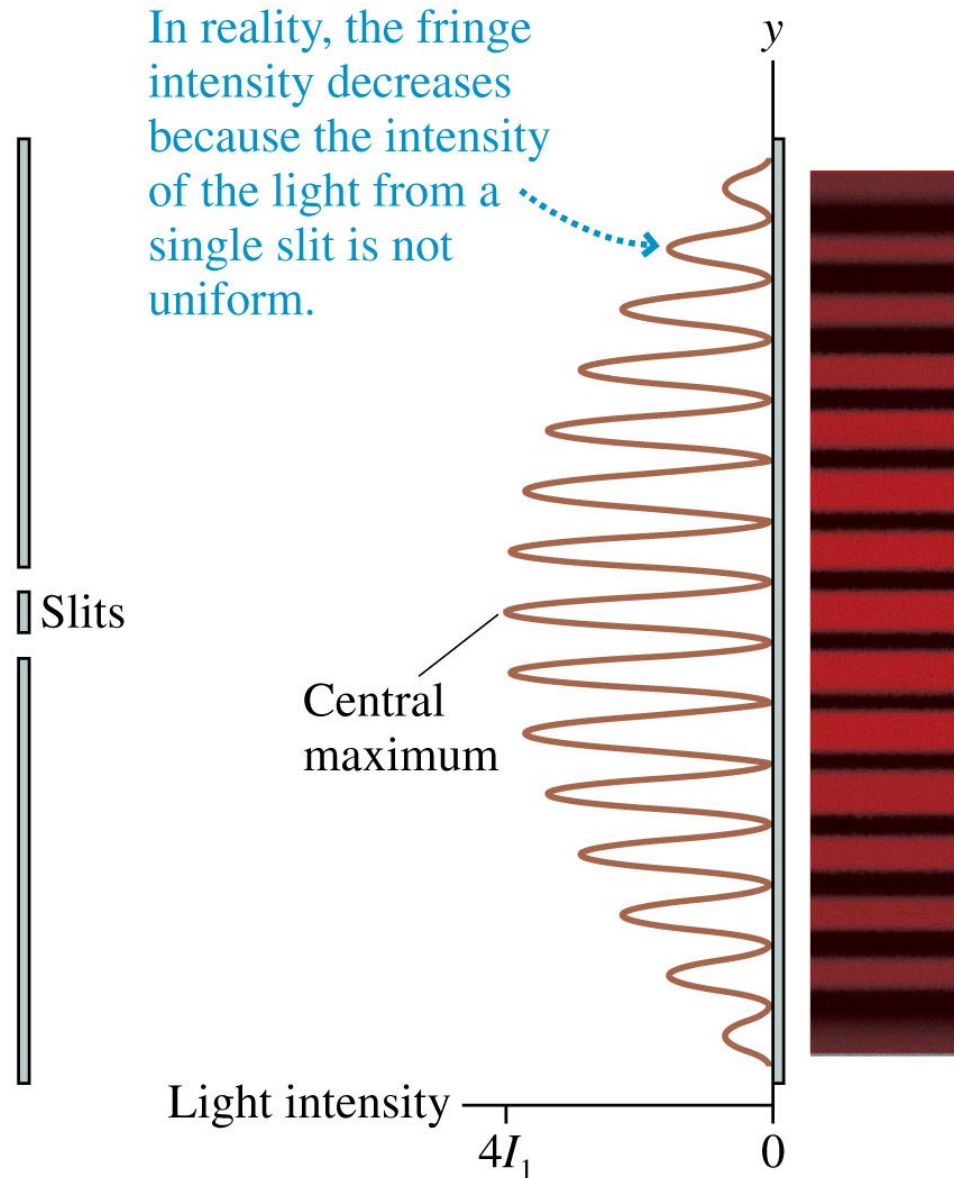
Intensity of the Double-Slit Interference Pattern



The intensity of the double-slit interference pattern at position y is:

$$I_{\text{double}} = 4I_1 \cos^2\left(\frac{\pi d}{\lambda L} y\right)$$

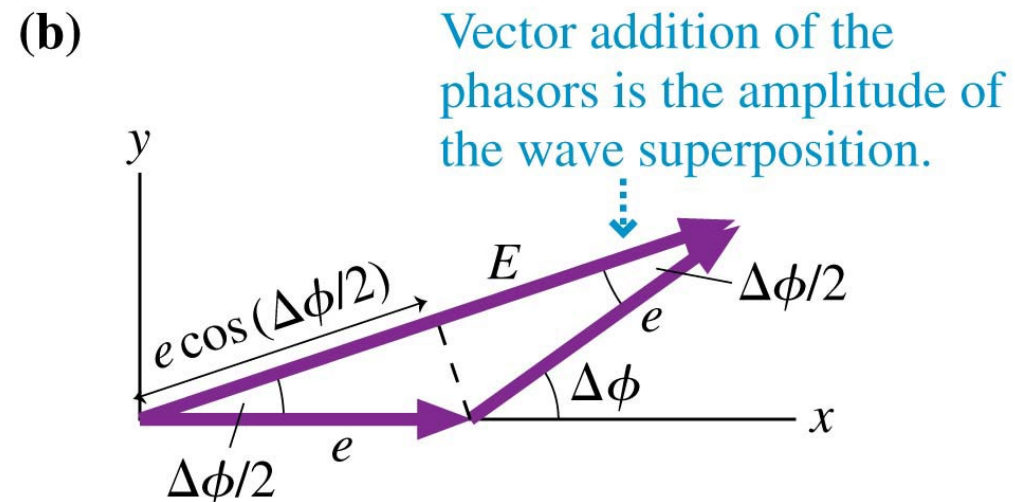
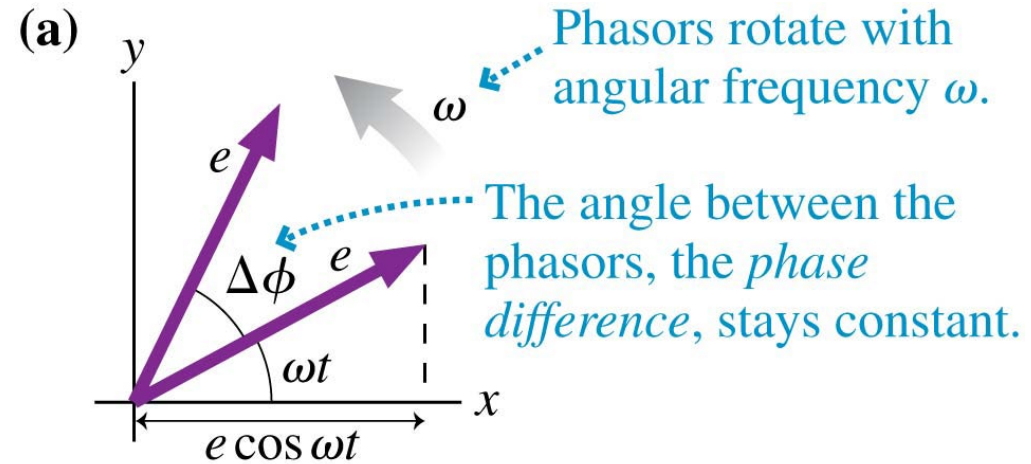
Intensity of the Double-Slit Interference Pattern



The actual intensity from a double-slit experiment slowly decreases as $|y|$ increases.

Phasors: A Closer Look at Superposition

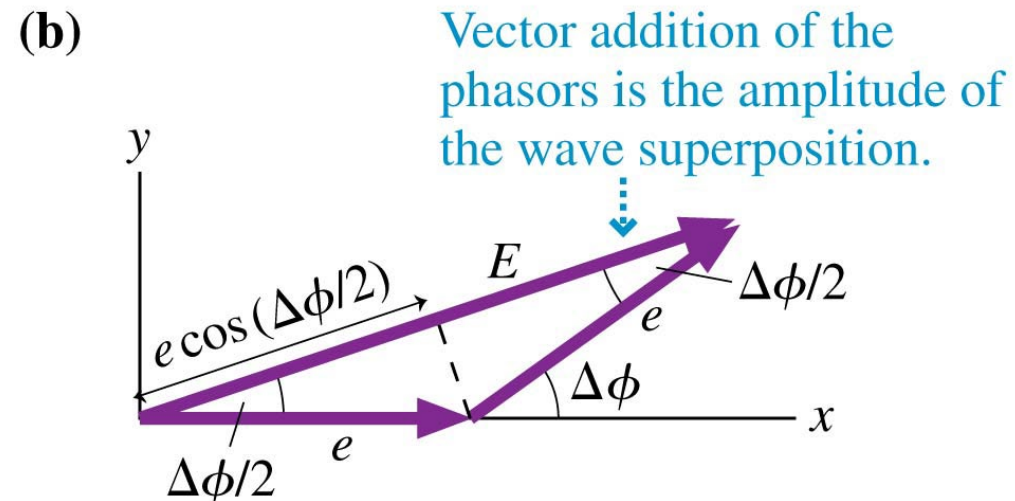
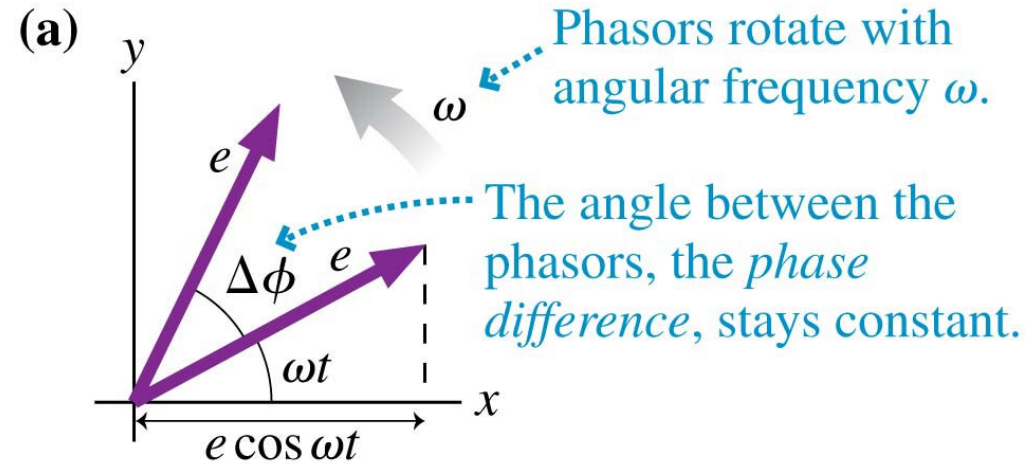
- To analyze double-slit interference, we can use a *phasor diagram* to show the addition of two electric field vectors.
- The projections of the phasors onto the x -axis are what we would add in a superposition calculation.



Phasors: A Closer Look at Superposition

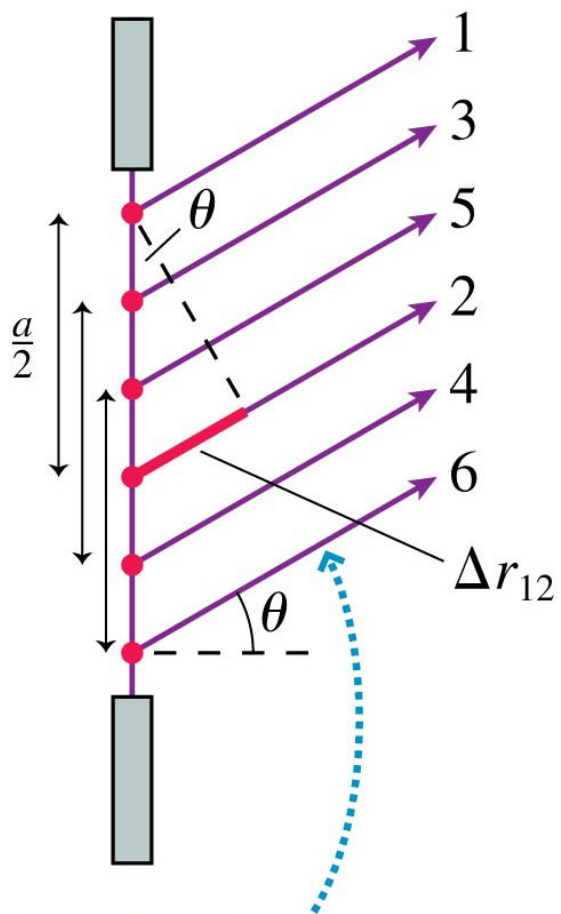
- If we add the phasors as vectors, using the tip-to-tail method, **the magnitude E of their vector sum is the electric field amplitude of the superposition of the two waves:**

$$E = \left| 2e \cos\left(\frac{\Delta\phi}{2}\right) \right|$$



Analyzing Single-Slit Diffraction

Each point on the wave front is paired with another point distance $a/2$ away.

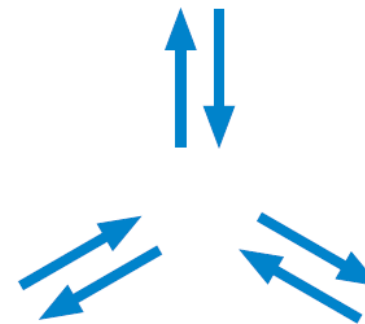


These wavelets all meet on the screen at angle θ . Wavelet 2 travels distance $\Delta r_{12} = (a/2)\sin\theta$ farther than wavelet 1.

- In this figure, wavelets 1 and 2 start from points that are $a/2$ apart.
- Every point on the wave front can be paired with another point distance $a/2$ away.
- If the path-length difference is $\Delta r = \lambda/2$, the wavelets arrive at the screen out of phase and interfere *destructively*.

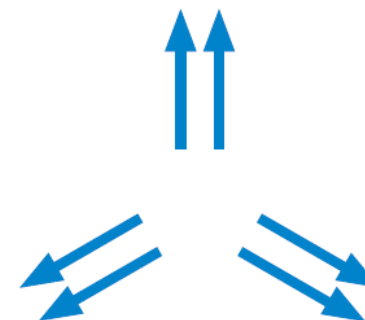
FIGURE 22.13 Destructive interference by pairs leads to net destructive interference, but constructive interference by pairs does *not* necessarily lead to net constructive interference.

(a)



Each pair of vectors interferes destructively. The vector sum of all six vectors is zero.

(b)

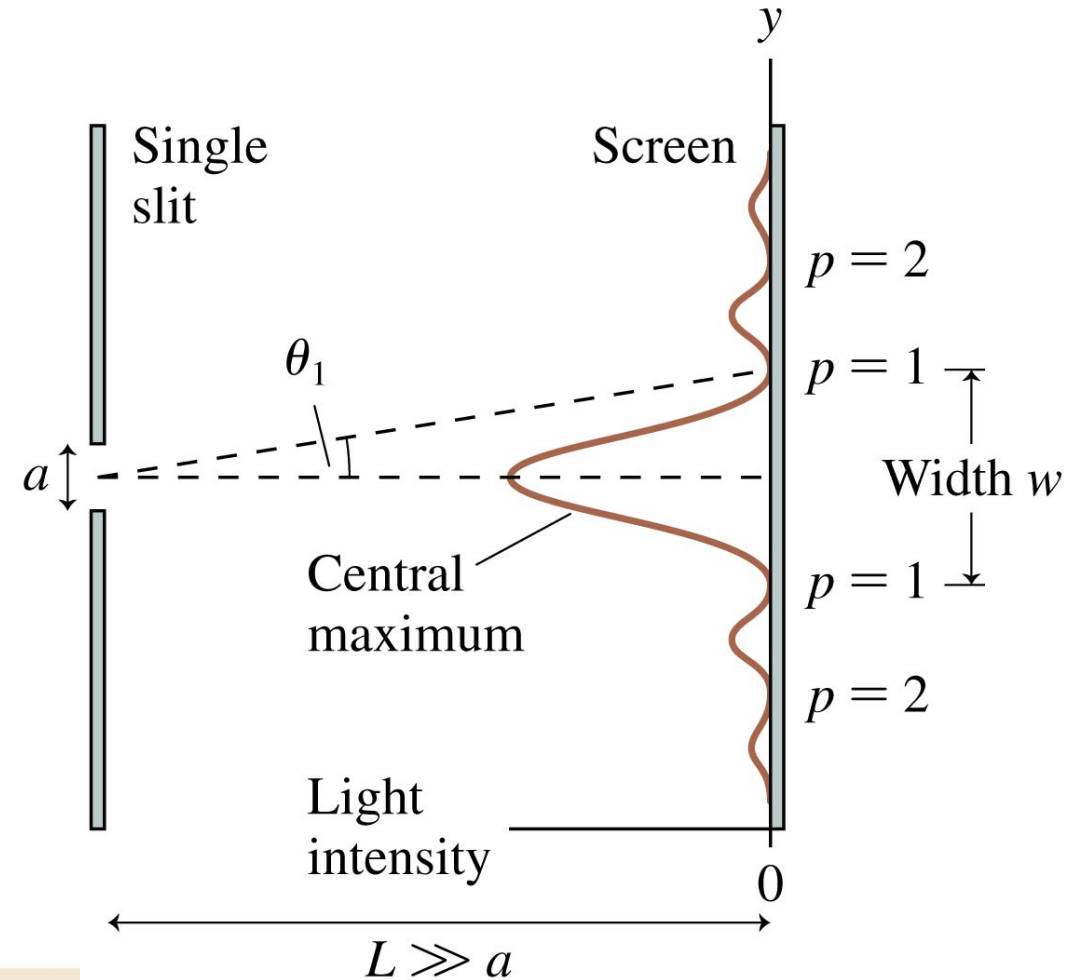


Each pair of vectors interferes constructively. Even so, the vector sum of all six vectors is zero.

Single-Slit Diffraction

- The light pattern from a single slit consists of a *central maximum* flanked by a series of weaker **secondary maxima** and dark fringes.
- The dark fringes occur at angles:

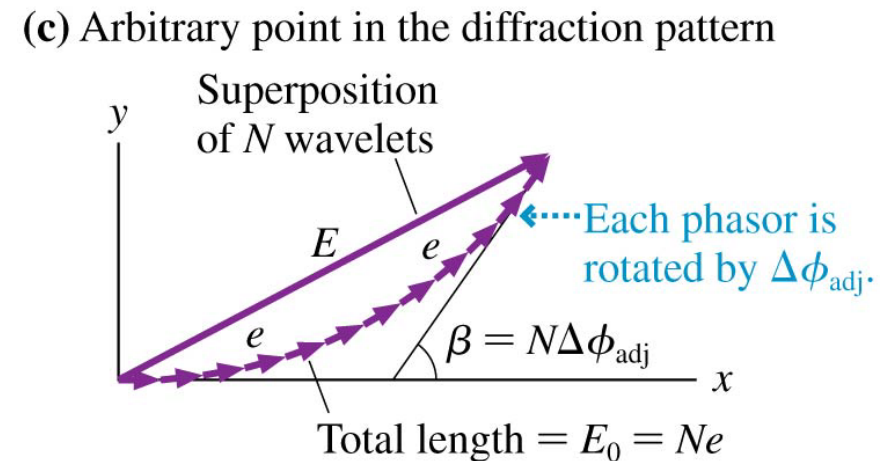
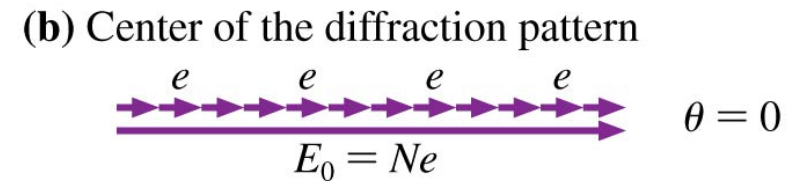
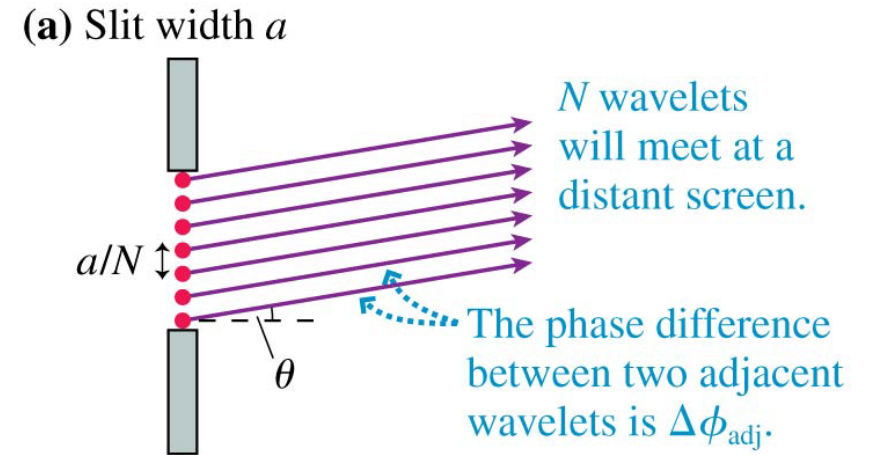
$$\theta_p = p \frac{\lambda}{a} \quad p = 1, 2, 3, \dots \quad (\text{angles of dark fringes})$$



Phasors: A Closer Look at Single-Slit Diffraction

- Figure (a) shows a slit of width a with N point sources of Huygens' wavelets, each separated by distance a/N .
- Figure (b) analyzes the diffraction at the center of the diffraction pattern.
- Figure (c) is the phasor diagram for superposition at an arbitrary point on the screen.
- The angle of the *last phasor* is:

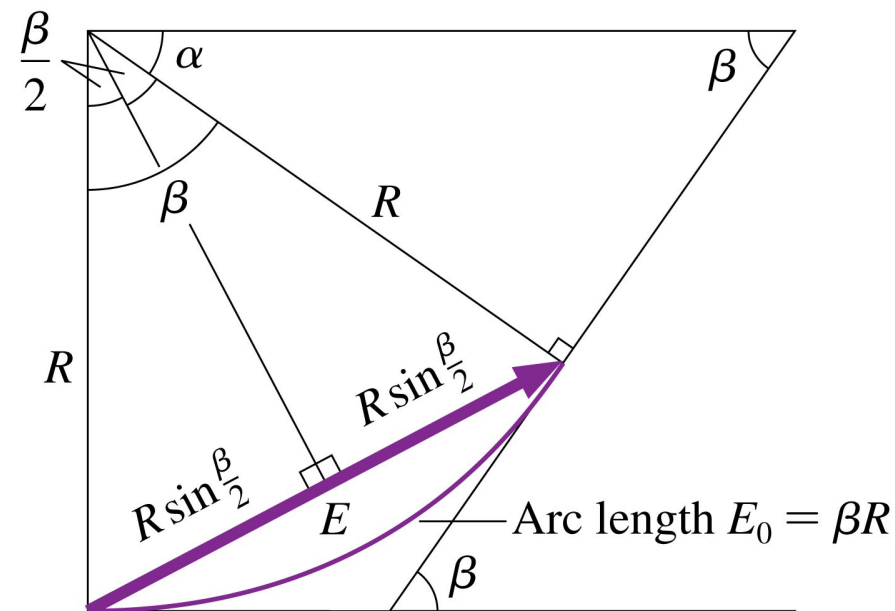
$$\beta = N \Delta\phi_{\text{adj}} = \frac{2\pi a \sin \theta}{\lambda}$$



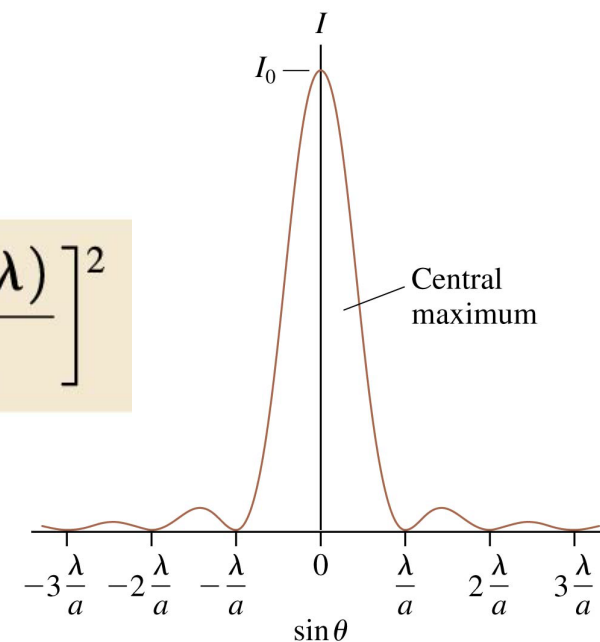
Phasors: A Closer Look at Single-Slit Diffraction

- E is the amplitude of the superposition of the N wavelets.
- To determine E , let $N \rightarrow \infty$.
- This makes our calculation exact, and it also makes our calculation easier, because now the chain of phasors, with length E_0 , is simply the arc of a circle.
- From the geometry, the angle subtending this arc must be β .
- The **amplitude** and **intensity** of the superposition are:

$$E = E_0 \frac{\sin(\beta/2)}{\beta/2} = E_0 \frac{\sin(\pi a \sin \theta / \lambda)}{\pi a \sin \theta / \lambda}$$



$$I_{\text{slit}} = I_0 \left[\frac{\sin(\pi a \sin \theta / \lambda)}{\pi a \sin \theta / \lambda} \right]^2$$

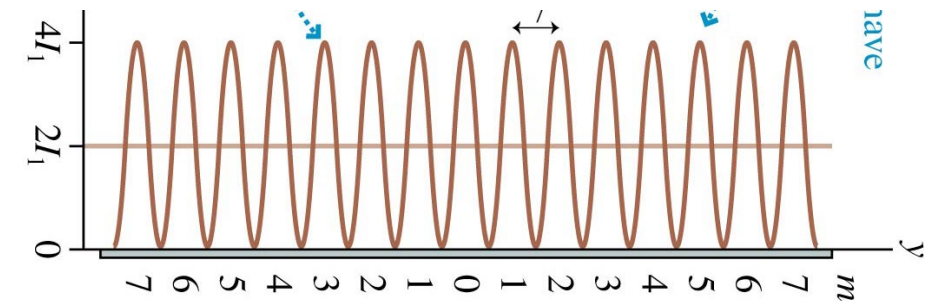
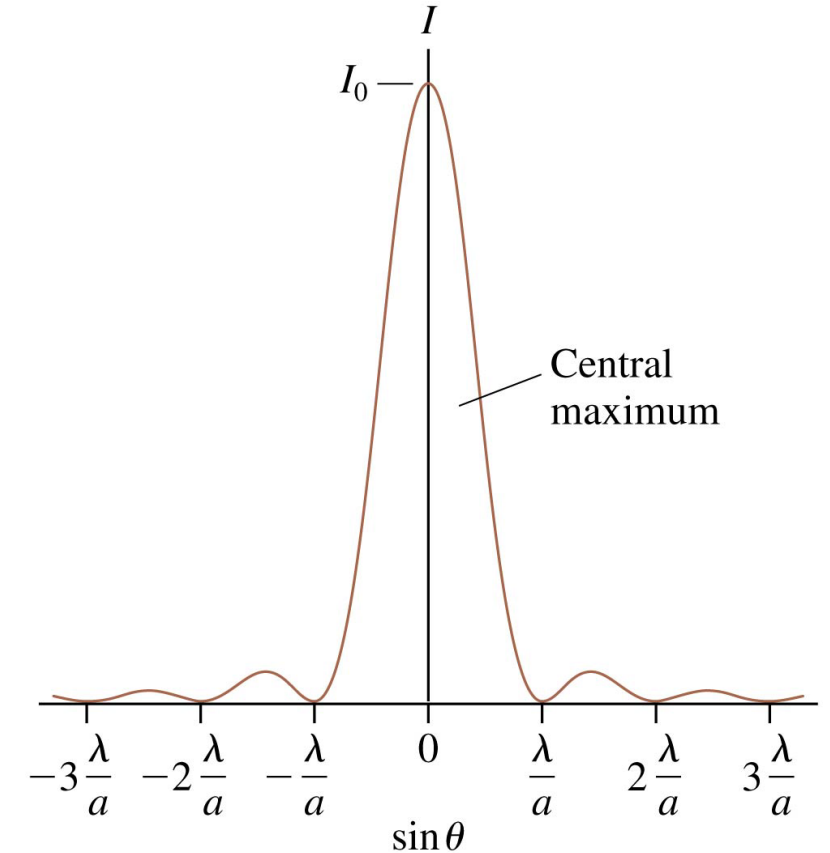


Phasors: The Complete Double-Slit Intensity

- We earlier calculated the ideal double-slit intensity for two slits separated by distance d .
- But each slit has width a , so the double-slit pattern is modulated by the single-slit diffraction intensity for a slit of width a .
- A realistic double-slit intensity, for small angles, is:

$$I_{\text{double}} = I_0 \left[\frac{\sin(\pi a y / \lambda L)}{\pi a y / \lambda L} \right]^2 \cos^2(\pi d y / \lambda L)$$

- The cosine term produces the fringe oscillations, but now the overall intensity is determined by the diffraction of the individual slits.



Phasors: A Closer Look at Diffraction

Single slit: 0.055 mm width

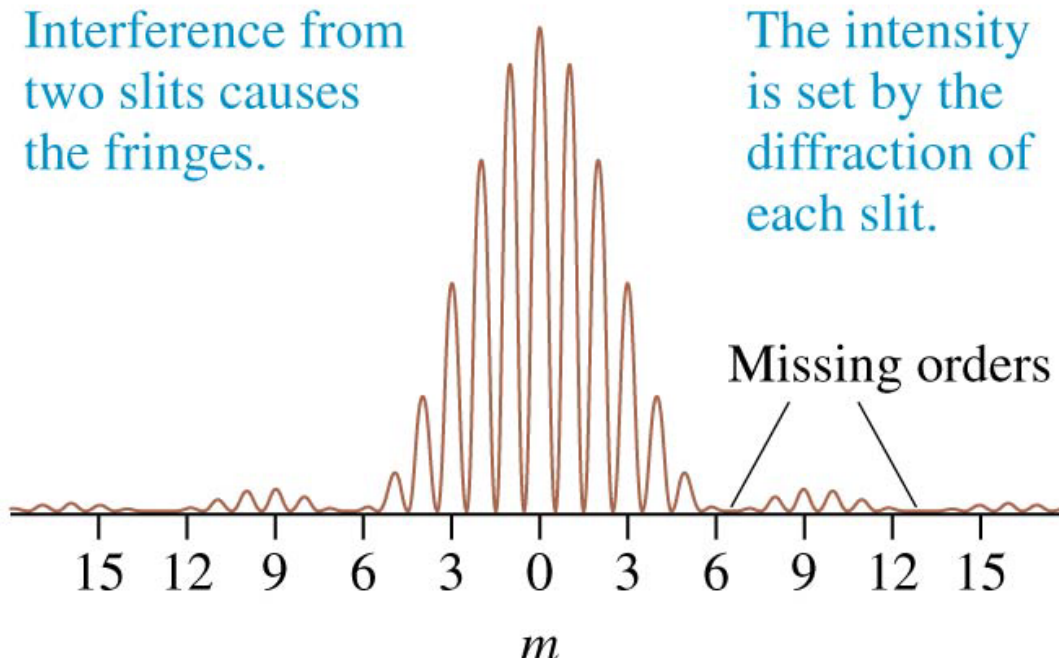


Two slits: 0.055 mm width, 0.35 mm separation



Interference from two slits causes the fringes.

The intensity is set by the diffraction of each slit.



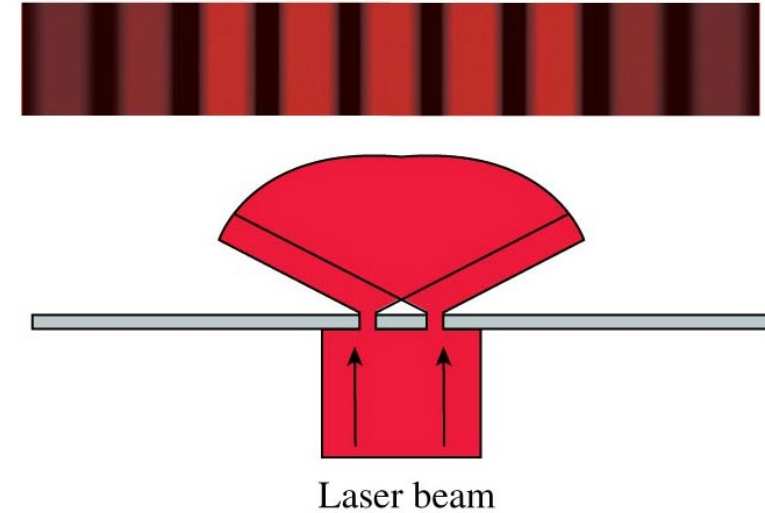
- If an interference maximum falls exactly on a minimum (a zero) in the single-slit diffraction pattern, then we have what is called a **missing order**:

$$m_{\text{missing}} = p \frac{d}{a} \quad p = 1, 2, 3, \dots$$

Q 22.10

A laboratory experiment produces a double-slit interference pattern on a screen. If the left slit is blocked, the screen will look like:

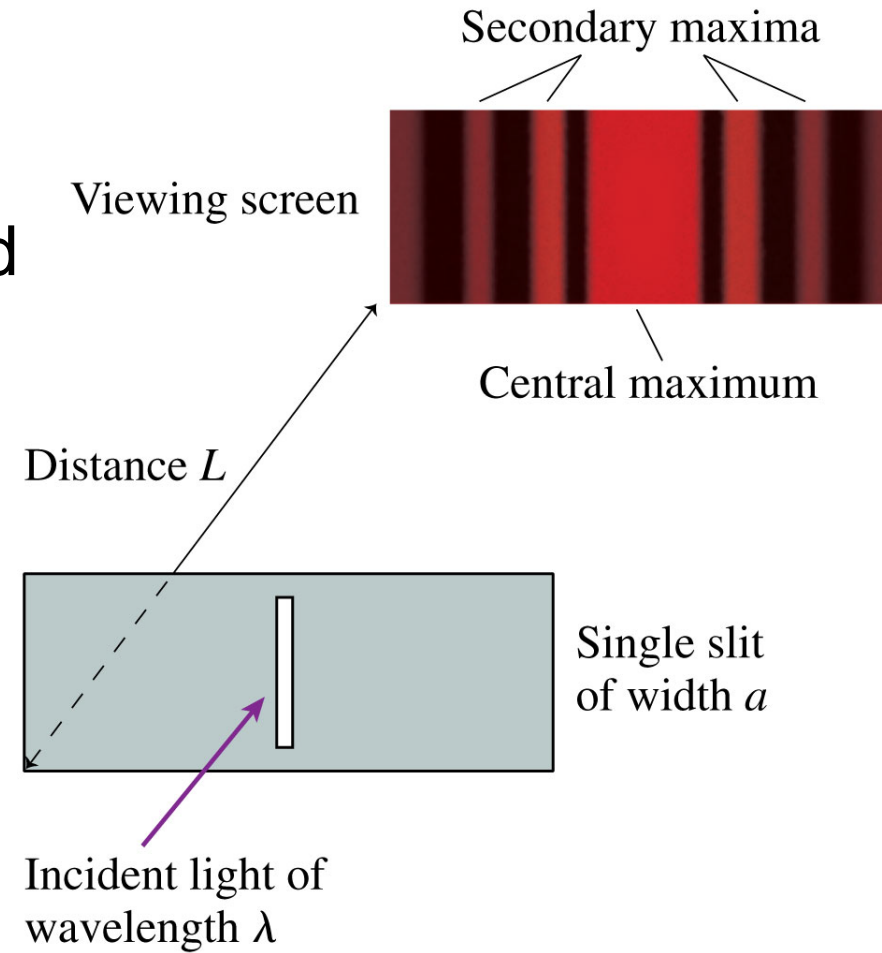
- A. 
- B. 
- C. 
- D. 



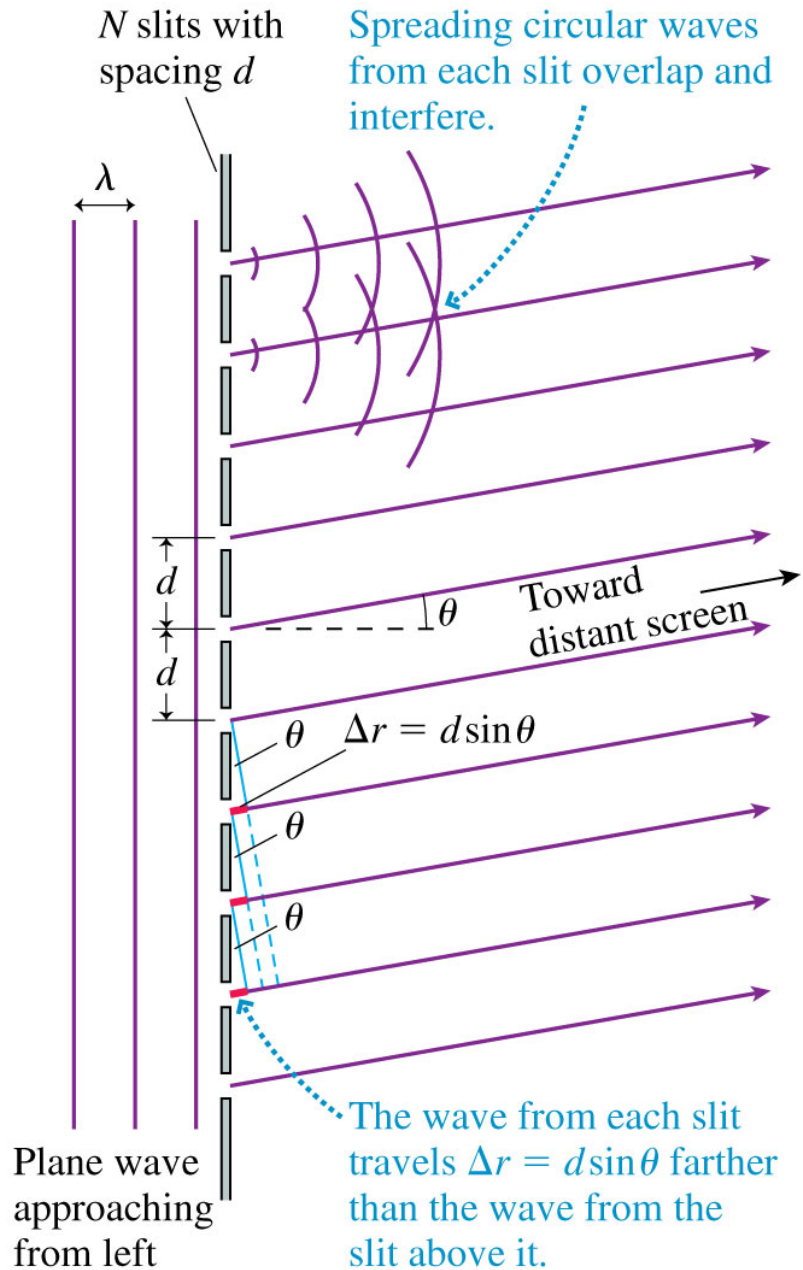
Q&A 22.10

A laboratory experiment produces a double-slit interference pattern on a screen. If the left slit is blocked the screen will look like:

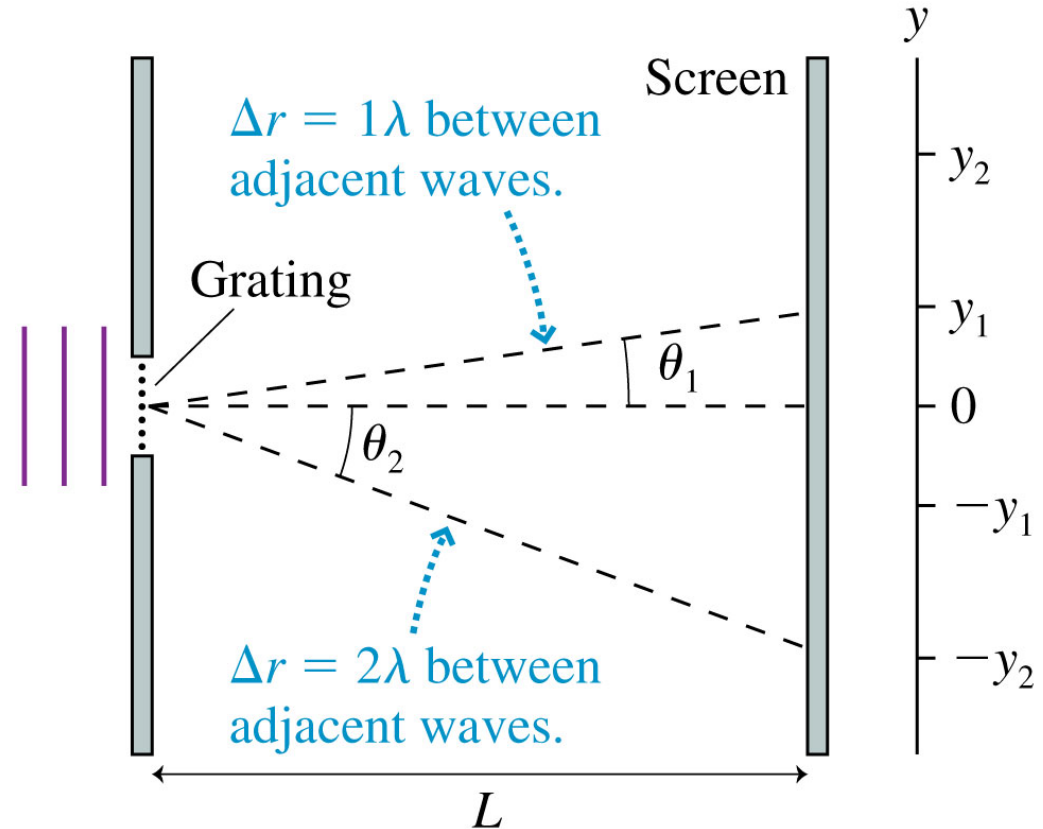
- A. 
- B. 
- C. 
- ✓ D. 



The Diffraction Grating



$$d \sin \theta_m = m\lambda \quad m = 0, 1, 2, 3, \dots$$



$$y_m = L \tan \theta_m \quad (\text{positions of bright fringes})$$

Q&A 22.8

In a laboratory experiment, a diffraction grating produces an interference pattern on a screen. If the number of slits in the grating is increased, with everything else (including the slit spacing) the same, then:

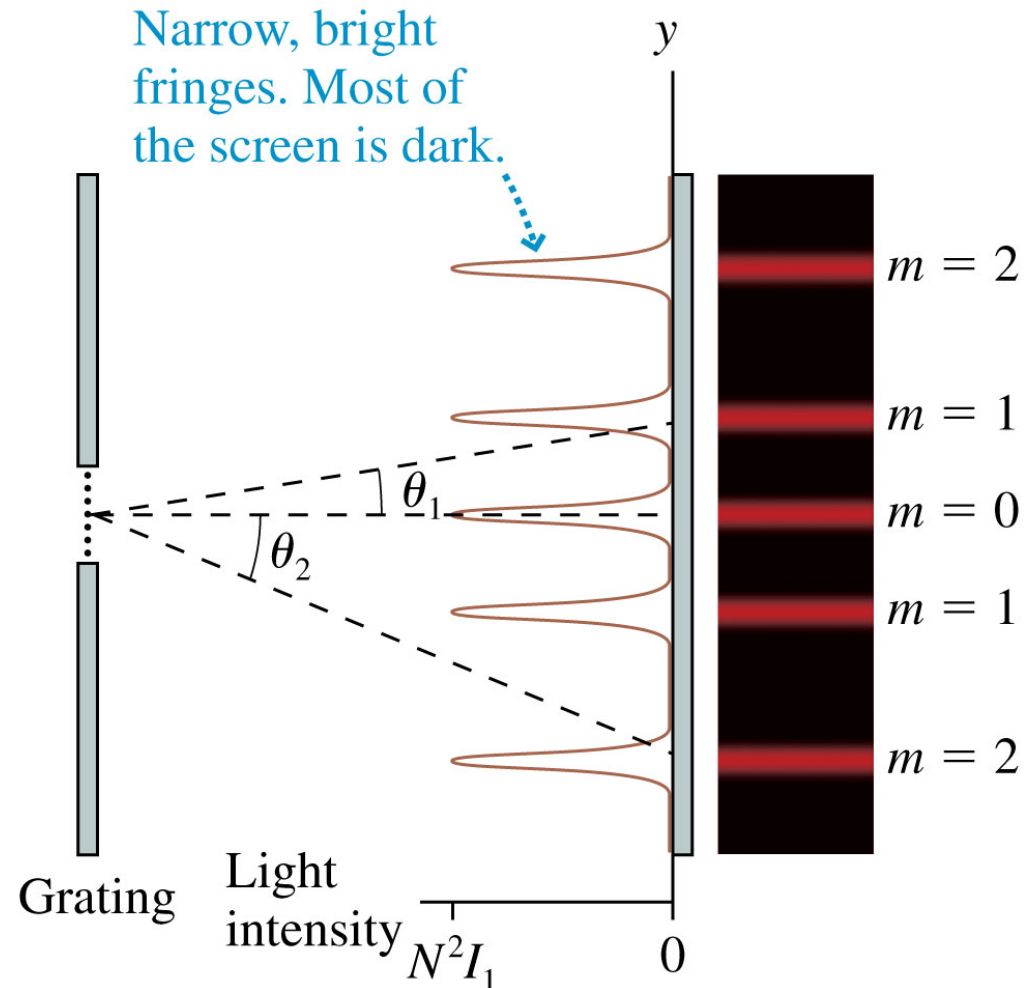


- A. The fringes stay the same brightness and get closer together.
- B. The fringes stay the same brightness and get farther apart.
- ✓ C. The fringes stay in the same positions but get brighter and narrower.
- D. The fringes stay in the same positions but get dimmer and wider.
- E. The fringes get brighter, narrower, and closer together.

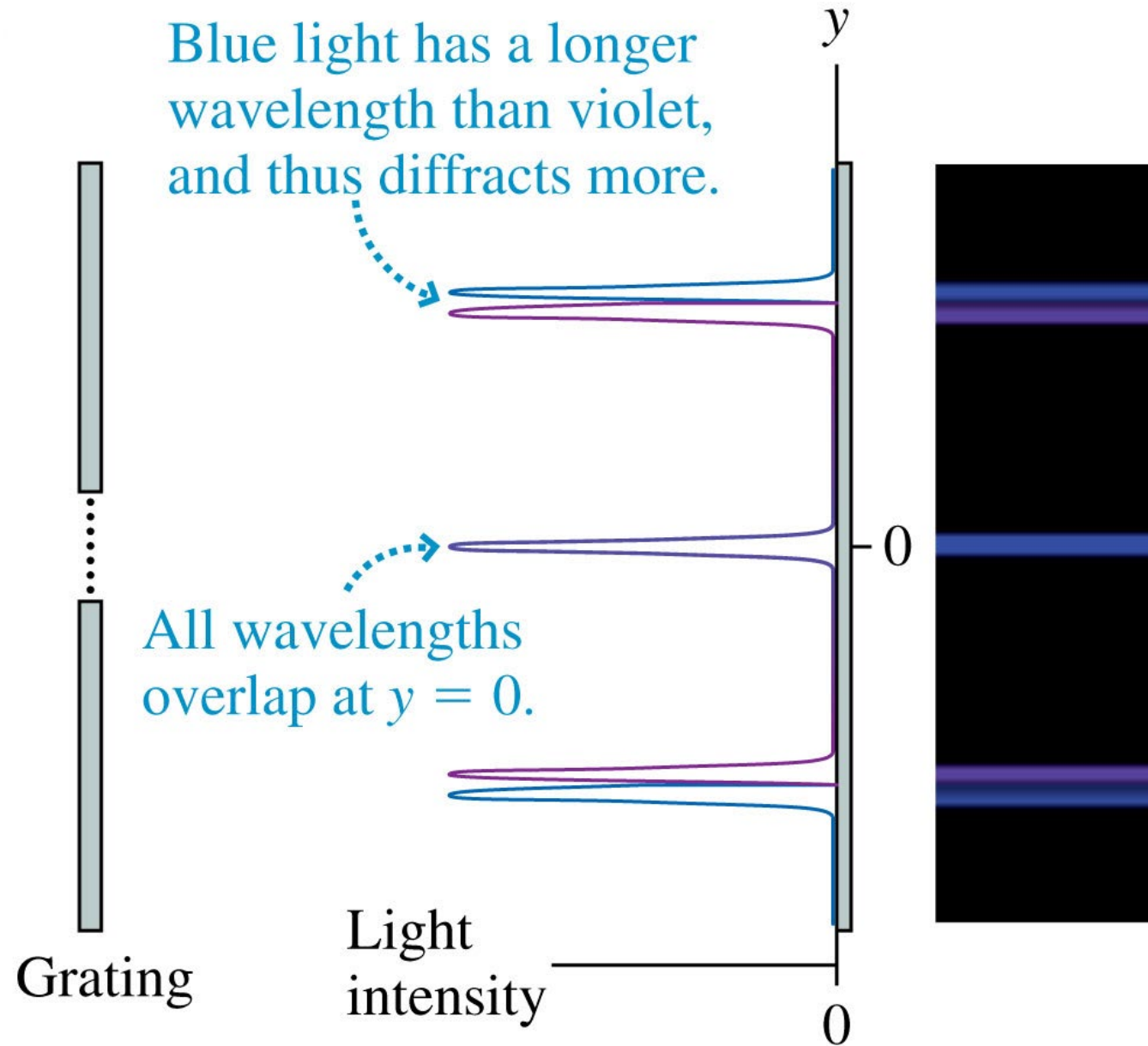
The Diffraction Grating

- The integer m is called the **order** of the diffraction.
- The wave **amplitude** at the points of constructive interference is Na .
- Because **intensity** depends on the square of the amplitude, the intensities of the bright fringes are:

$$I_{\max} = N^2 I_1$$



The Diffraction Grating

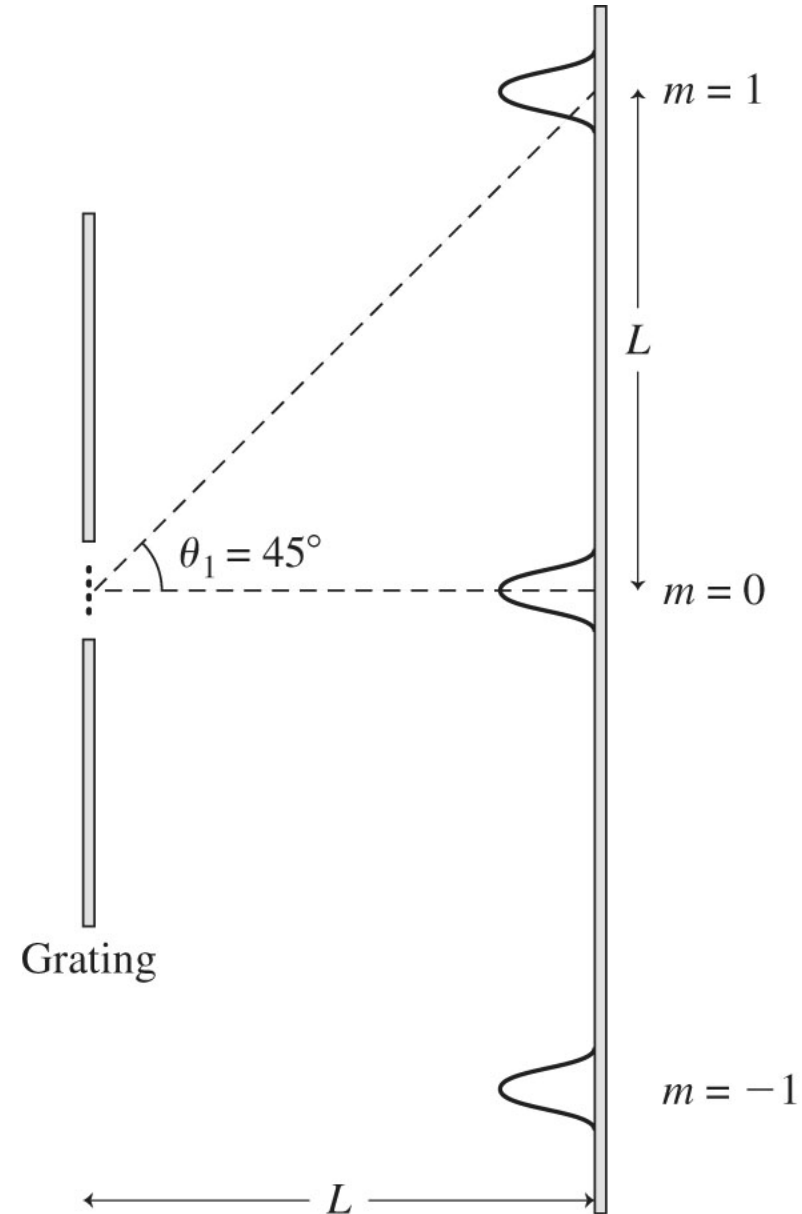


Example: Problem 22.52

A diffraction grating has slit spacing d . Fringes are viewed on a screen at distance L .

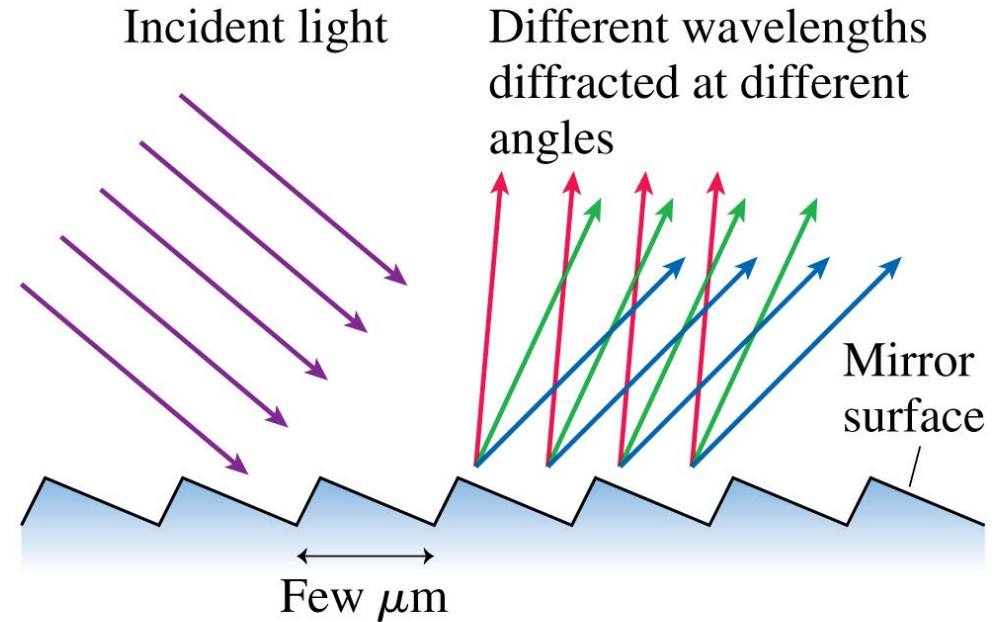
→ What wavelength of light produces a first-order fringe on the viewing screen at distance L from the center of the screen?

$$\lambda = \frac{d}{\sqrt{2}}$$



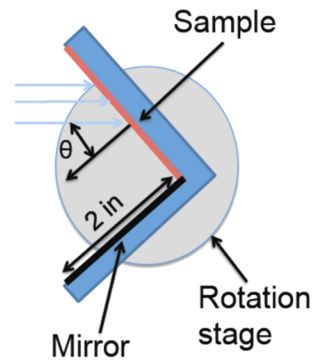
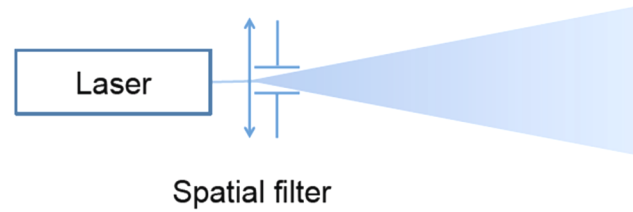
Reflection Gratings

- In practice, most diffraction gratings are manufactured as **reflection gratings**.
- The interference pattern is exactly the same as the interference pattern of light transmitted through N parallel slits.

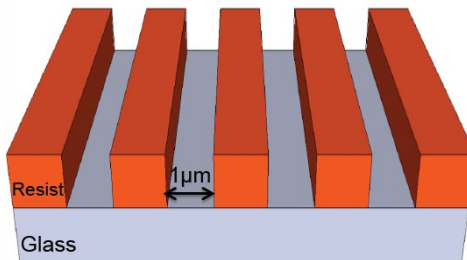
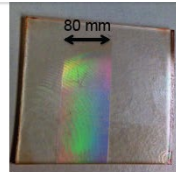


A reflection grating can be made by cutting parallel grooves in a mirror surface. These can be very precise, for scientific use, or mass produced in plastic.

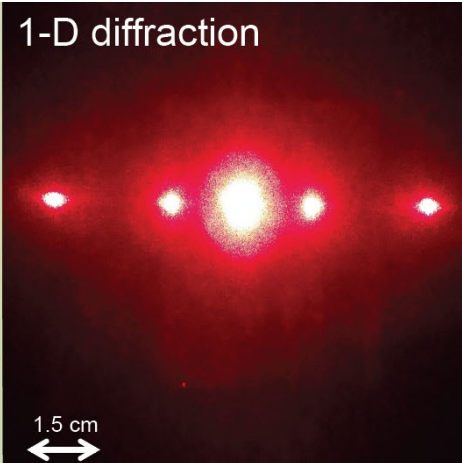
Lloyd's Mirror Interferometer



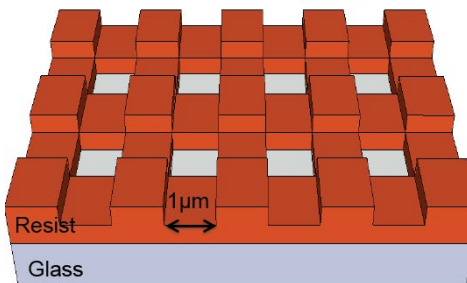
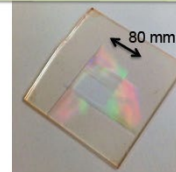
1-D structure



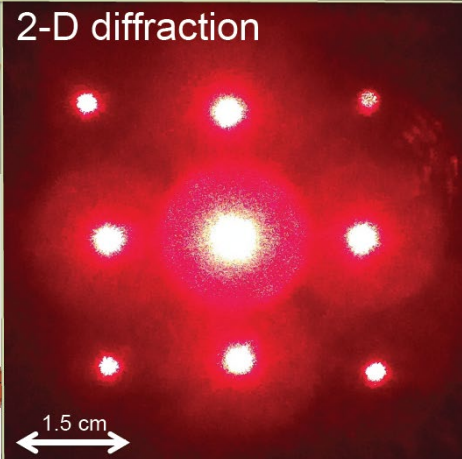
1-D diffraction



2-D structure

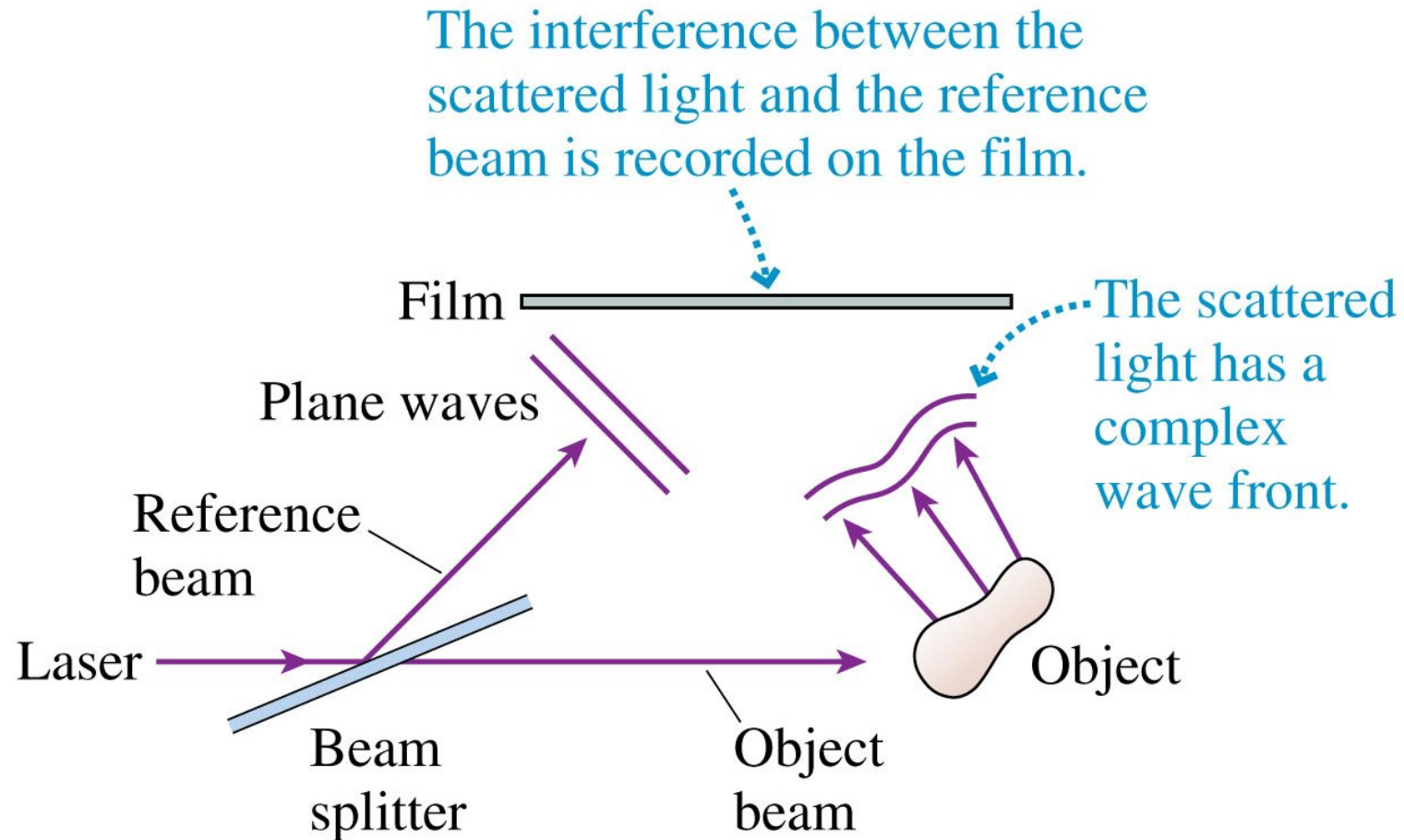


2-D diffraction



Holography

The figure shows shows how a **hologram** is recorded.



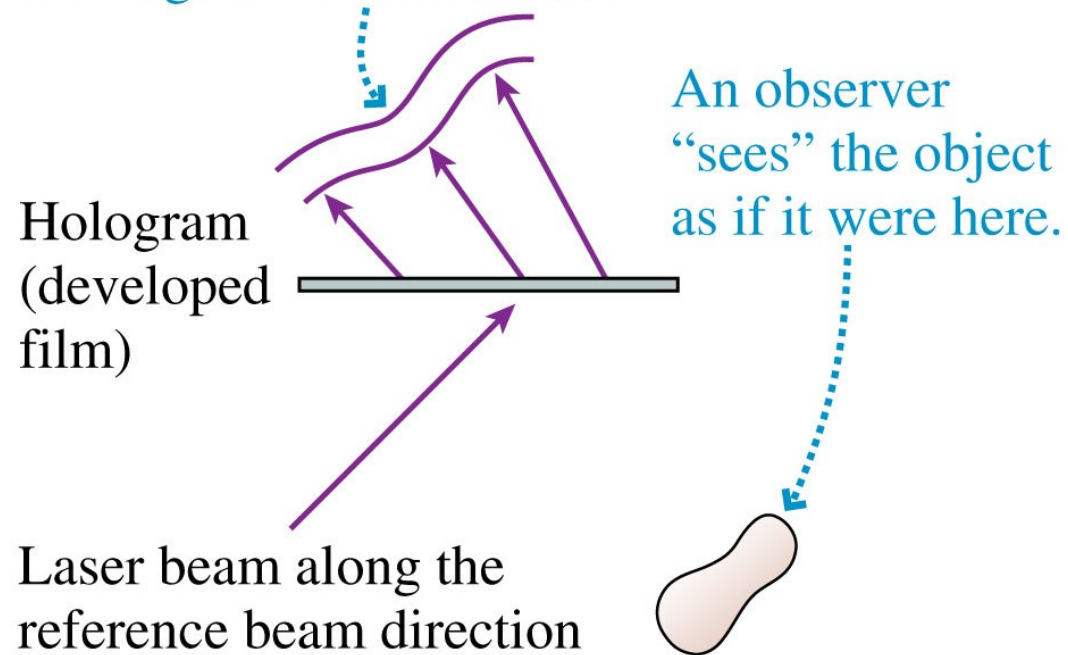
Hologram



Holography

The hologram is “played” by sending just the reference beam through it.

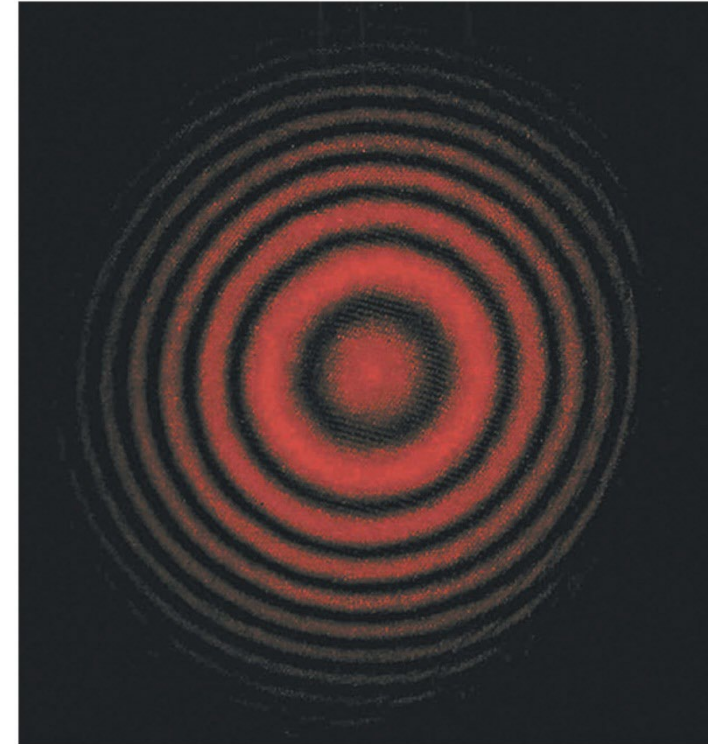
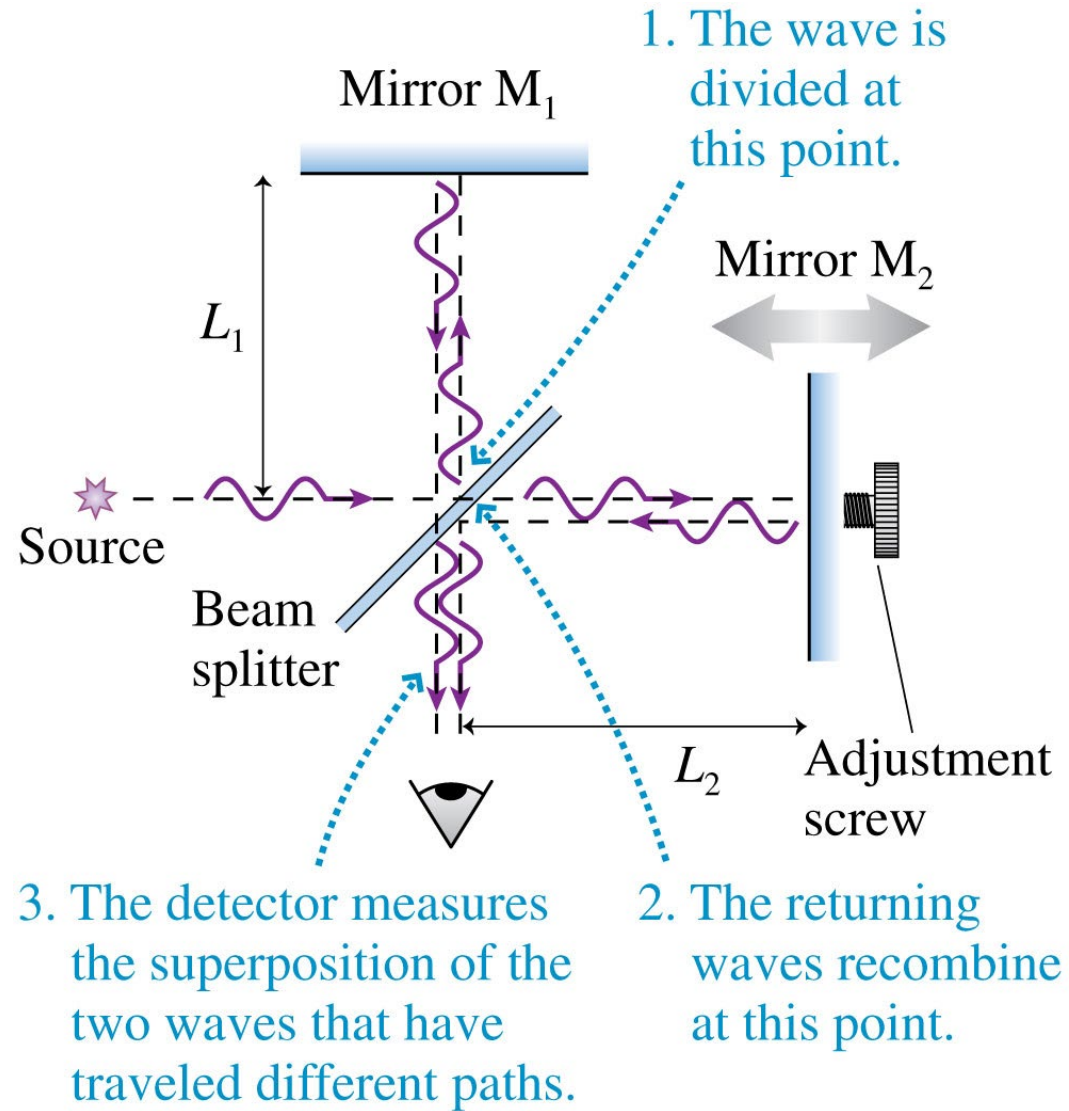
The diffraction of the laser beam through the light and dark patches of the film reconstructs the original scattered wave.



The view is three dimensional because, by moving your head with respect to the hologram, you can see different portions of the wave front.



The Michelson Interferometer



- The number Δm of central maxima appearing as M_2 moves through a distance ΔL_2 is:

$$\Delta m = \frac{\Delta L_2}{\lambda/2}$$

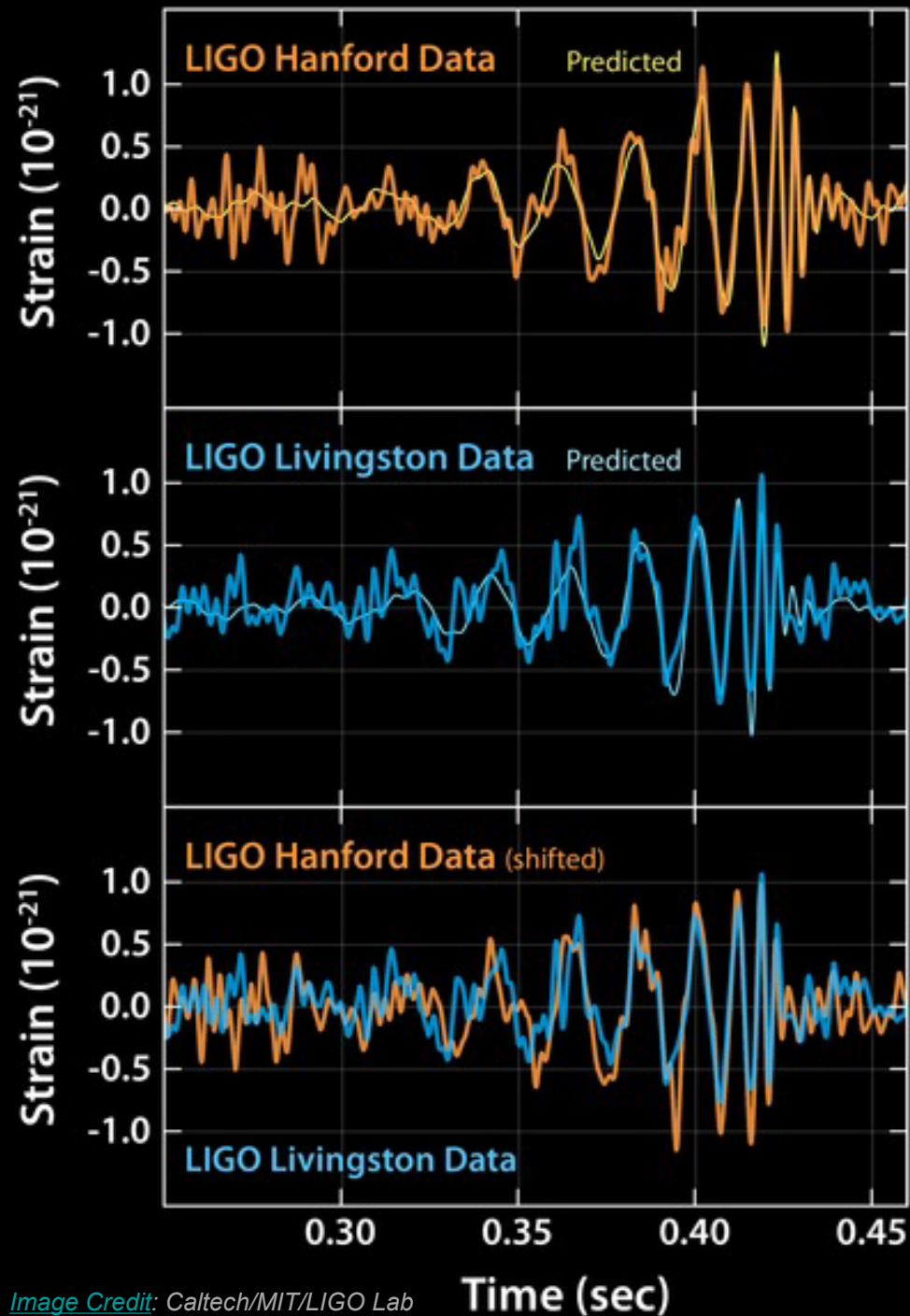
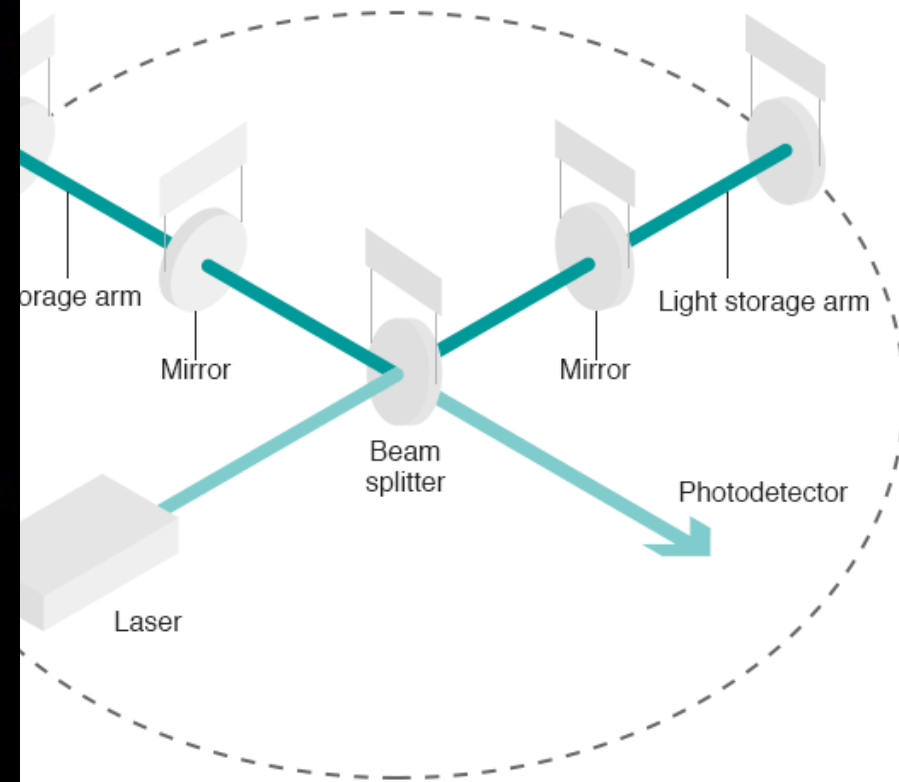


Image Credit: Caltech/MIT/LIGO Lab

Interferometer: How a gravitational wave hunter works



Gravitational waves alternately stretch and squeeze the space they pass through

